

18. Data Transfer Controller (DTC)

18.1 Overview

A Data Transfer Controller (DTC) module is provided for transferring data when activated by an interrupt request.

Table 18.1 lists the DTC specifications and Figure 18.1 shows DTC block diagram.

Table 18.1 DTC specifications

Item	DTC0 (for CPU0)/DTC1 (for CPU1)
Transfer modes	- Normal transfer mode A single activation leads to a single data transfer. - Repeat transfer mode A single activation leads to a single data transfer. The transfer address returns to the start address after the number of data transfers reaches the specified repeat size. The maximum number of repeat transfers is 256 and the maximum data transfer size is 256×64 bits (2048 bytes). - Block transfer mode A single activation leads to a transfer of a single block. The maximum block size is 256×64 bits = 2048 bytes.
Transfer channel	- Channel transfer corresponding to the interrupt source is possible (transferred by a DTC activation request from the ICU). - Multiple data units can be transferred on a single activation source (chain transfer). - Chain transfers are selectable to either execute when the counter is 0 or always execute.
Transfer space	4 GB area from 0x0000_0000 to 0xFFFF_FFFF excluding reserved areas.
Data transfer units	- Single data unit: 1 byte (8 bits), 1 halfword (16 bits), 1 word (32 bits), 1 doubleword (64 bits). - Single block size: 1 to 256 data units.
CPU interrupt source	- An interrupt request can be generated to the CPU on a DTC activation interrupt - An interrupt request can be generated to the CPU after a single data transfer - An interrupt request can be generated to the CPU after a data transfer of a specified volume.
Processing on DTC transfer error	- When the DTC transfer error occurs, it stops the transfer that caused the error - Request to clear the register for activation request of DTC error number to ICU
Error response detection interrupt	Generated when the DTC transfer error occurs
Event link function	Generated after one data transfer (for block, after one block transfer)
Read skip	Read of transfer information can be skipped
Write-back skip	When the transfer source or destination address is specified as fixed, a write-back of transfer information can be skipped
Module-stop function	Module-stop state can be set to reduce power consumption
TrustZone Filter	Security attribution and privilege attribution can be set for each activation source.

Note: Security attribution Register of DTC is described in ICU.ICUSARG, ICU.ICUSARH, ICU.ICUSARI, ICU.ICUSARJ, ICU.ICUSARK and ICU.ICUSARL

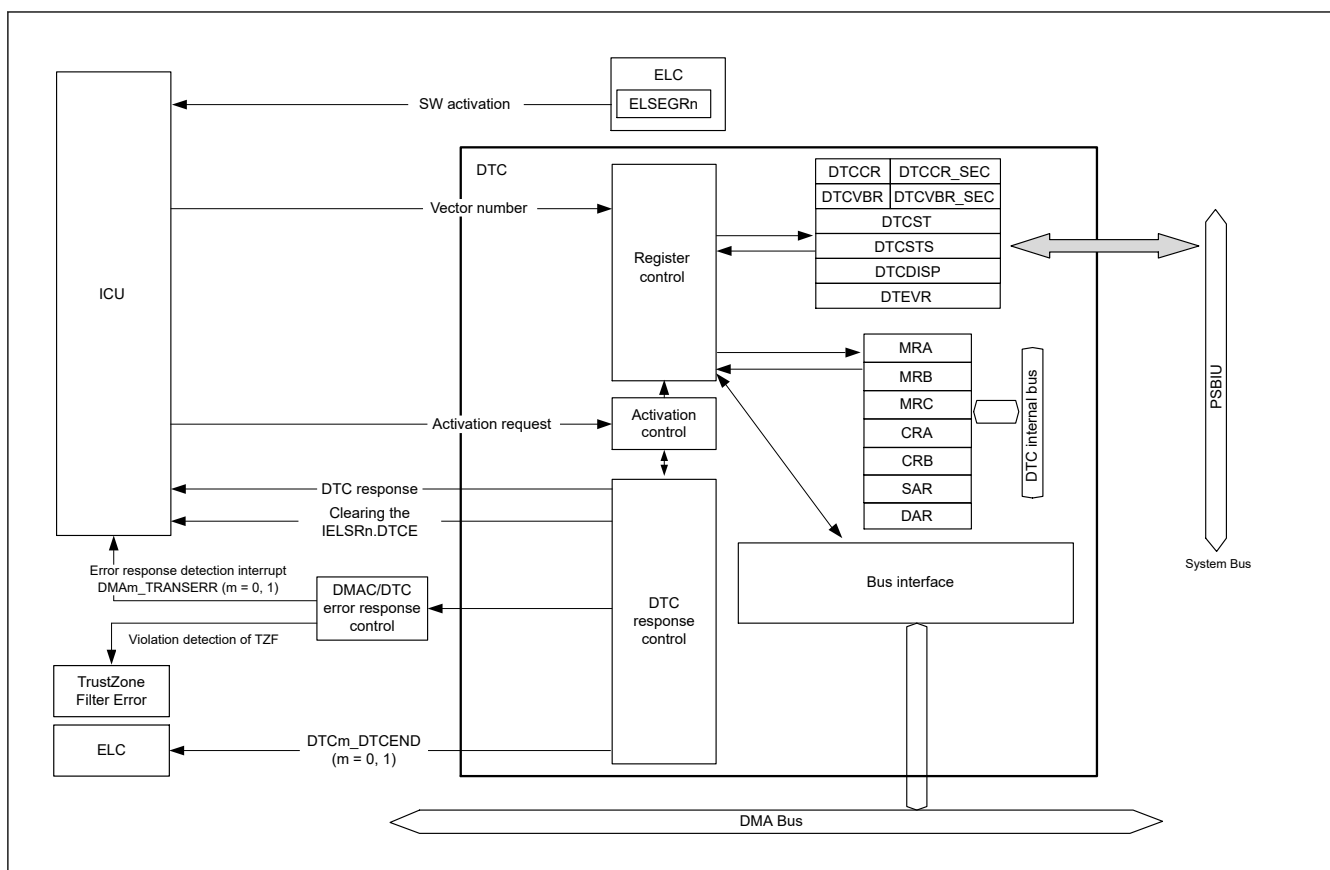


Figure 18.1 DTC block diagram

See [section 14.1. Overview](#) in [section 14, Interrupt Controller Unit \(ICU\)](#) for the connections between the DTC and NVIC in the CPU.

18.2 Register Descriptions

MRA, MRB, MRC, SAR, DAR, CRA, and CRB are all DTC internal registers that cannot be directly accessed from the CPU. Values to be set in these DTC internal registers are placed in the SRAM area as transfer information. When an activation request is generated, the DTC reads the transfer information from the SRAM area and sets it in its internal registers. After the data transfer ends, the internal register contents are written back to the SRAM area as transfer information.

The DTC0 and DTC1 registers have the same base address. DTC0 is dedicated to CPU0 and can only be accessed from CPU0. DTC1 is dedicated to CPU1 and can only be accessed from CPU1.

18.2.1 DTCSAR : DTC Controller Security Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x30

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DTCS TSA1
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DTCS TSA0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DTCSTSA0	DTC0 Security Attribution 0: Secure 1: Non-secure	R/W
15:1	—	These bits are read as 0. The write value should be 0.	R/W
16	DTCSTSA1	DTC1 Security Attribution 0: Secure 1: Non-secure	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

This register sets the DTCST and DTCSTS security attribution.

DTCSTSA0 bit (DTC0 Security Attribution)

This bit specifies security attribution of registers for DTC0.DTCST and DTC0.DTCSTS.

Do not write to the DTCSTSA0 bit while DTC0 transfer is enabled or a bus master is writing to the DTC0 registers.

DTCSTSA1 bit (DTC1 Security Attribution)

This bit specifies security attribution of registers for DTC1.DTCST and DTC1.DTCSTS.

Do not write to the DTCSTSA1 bit while DTC1 transfer is enabled or a bus master is writing to the DTC1 registers.

18.2.2 MRA : DTC Mode Register A

Base address: DTCVBR_SEC
DTCVBR

Offset address: $0x03 + 0x4 \times \text{Vector number}$
(Inaccessible directly from the CPU. See [section 18.3.1. Allocating Transfer Information and DTC Vector Table.](#))

Bit position:	7	6	5	4	3	2	1	0
Bit field:	MD[1:0]	SZ[1:0]	SM[1:0]	—	WBDI	S		

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
0	WBDIS	Write Back Disable 0: Write back to the DTC transfer information at the end of DTC data transfer. 1: Do not write back to the DTC transfer information at the end of DTC data transfer.	—
1	—	The read value is undefined. The write value should be 0.	—
3:2	SM[1:0]	Transfer Source Address Addressing Mode 0 0: Address in the SAR register is fixed (write-back to SAR is skipped.) 0 1: Address in the SAR register is fixed (write-back to SAR is skipped.) 1 0: SAR value is incremented after data transfer: +1 when SZ[1:0] = 00b +2 when SZ[1:0] = 01b +4 when SZ[1:0] = 10b +8 when SZ[1:0] = 11b 1 1: SAR value is decremented after data transfer: -1 when SZ[1:0] = 00b -2 when SZ[1:0] = 01b -4 when SZ[1:0] = 10b -8 when SZ[1:0] = 11b	—
5:4	SZ[1:0]	DTC Data Transfer Size 0 0: Byte (8-bit) transfer 0 1: Halfword (16-bit) transfer 1 0: Word (32-bit) transfer 1 1: Doubleword (64-bit) transfer	—

Bit	Symbol	Function	R/W
7:6	MD[1:0]	DTC Transfer Mode Select 0 0: Normal transfer mode 0 1: Repeat transfer mode 1 0: Block transfer mode 1 1: Setting prohibited	—

The MRA register cannot be accessed directly from the CPU, however the CPU can access the SRAM area (transfer information (n) start address + 0x03) and DTC transfers it automatically to and from the MRA register. See [section 18.3.1. Allocating Transfer Information and DTC Vector Table](#).

WBDIS bit (Write Back Disable)

This bit selects whether to write back the transfer information at the end of data transfer.

When the WBDIS bit is 0, the updated DTC transfer information is written back.

When the WBDIS bit is 1, the transfer information is not written back even if the address is set to be incremented after transfer, and the same data transfer is performed for each start request. Since the transfer information is not written back, you can put the DTC transfer information in ROM. When the WBDIS bit is 1, the following operations are performed for each transfer mode.

When setting the MRC.DISPE bit to 1, also set the WBDIS bit to 1 (do not write back). If there is at least one transfer information with the WBDIS bit set to 1, set DTCCR.RRS to 0 (do not perform read skip). When used in secure state, access DTCCR_SEC.RRS instead of DTCCR.RRS.

(1) Normal transfer mode and repeat transfer mode

One startup factor transfers 1 byte, 1 halfword, 1 word, and 1 doubleword. The transfer address and the number of transfers are not updated, and the same transfer is repeated for each transfer request. Even if the number of transfers is 1, ICU.DTCERn.DTCE will not be cleared and DTC transfer will continue due to the next activation factor.

(2) Block transfer mode

One block is transferred by one activation factor. The transfer address and block transfer count are not updated, and the same block transfer is repeated for each transfer request. Even if the number of block transfers is 1, ICU.DTCERn.DTCE does not clear to 0 and continues DTC transfer due to the next activation factor.

18.2.3 MRB : DTC Mode Register B

Base address: DTCVBR_SEC
DTCVBR

Offset address: $0x02 + 0x4 \times \text{Vector number}$
(Inaccessible directly from the CPU. See [section 18.3.1. Allocating Transfer Information and DTC Vector Table](#).)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	CHNE	CHNS	DISEL	DTS	DM[1:0]	—	—	—

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
1:0	—	The read values are undefined. The write value should be 0.	—

Bit	Symbol	Function	R/W
3:2	DM[1:0]	Transfer Destination Address Addressing Mode 0 0: Address in the DAR register is fixed (write-back to DAR is skipped) 0 1: Address in the DAR register is fixed (write-back to DAR is skipped) 1 0: DAR value is incremented after data transfer: +1 when MRA.SZ[1:0] = 00b +2 when SZ[1:0] = 01b +4 when SZ[1:0] = 10b +8 when SZ[1:0] = 11b 1 1: DAR value is decremented after data transfer: -1 when MRA.SZ[1:0] = 00b -2 when SZ[1:0] = 01b -4 when SZ[1:0] = 10b -8 when SZ[1:0] = 11b	—
4	DTS	DTC Transfer Mode Select 0: Select transfer destination as repeat or block area. 1: Select transfer source as repeat or block area.	—
5	DISEL	DTC Interrupt Select 0: Generate an interrupt request to the CPU when specified data transfer is complete. 1: Generate an interrupt request to the CPU each time DTC data transfer is performed.	—
6	CHNS	DTC Chain Transfer Select 0: Chain transfer is continuous. 1: Chain transfer occurs only when the transfer counter changes from 1 to 0 or 1 to CRAH.	—
7	CHNE	DTC Chain Transfer Enable 0: Chain transfer is disabled. 1: Chain transfer is enabled.	—

The MRB register cannot be accessed directly from the CPU, however the CPU can access the SRAM area (transfer information (n) start address + 0x02) and DTC transfers it automatically to and from the MRB register. See [section 18.3.1. Allocating Transfer Information and DTC Vector Table](#).

DM[1:0] bits (Transfer Destination Address Addressing Mode)

The DM[1:0] bits are to fix the address of the DAR register or specify increment / decrement of the DAR register after transfer.

DTS bit (DTC Transfer Mode Select)

The DTS bit specifies whether the transfer source or destination is the repeat or block area in repeat or block transfer mode.

DISEL bit (DTC Interrupt Select)

The DISEL bit specifies the condition for generating an interrupt request to the CPU.

CHNS bit (DTC Chain Transfer Select)

The CHNS bit selects the chain transfer condition. When CHNE is 0, the CHNS setting is ignored. For details on the conditions for chain transfer, see [Table 18.3](#).

When the next transfer is chain transfer, completion of the specified number of transfers is not determined, the activation source flag is not cleared, and an interrupt request to the CPU is not generated.

CHNE bit (DTC Chain Transfer Enable)

The CHNE bit enables chain transfer. The chain transfer condition is selected by the CHNS bit. For details on chain transfer, see [section 18.4.6. Chain Transfer](#).

Note: When the chain transfer settings are CHNE = 1 and CHNS = 0, the transfer count according to the transfer settings will be -1 for each transfer request. Therefore, the transfer count for each transfer information must match.

The DAR sets the transfer destination start address and cannot be accessed directly from the CPU. However, the CPU can access the SRAM area (transfer information (n) start address + 0x08) and DTC transfers it automatically to and from the DAR register. See [section 18.3.1. Allocating Transfer Information and DTC Vector Table](#).

Misalignment is prohibited for DTC transfers. Bit[0] must be 0 when MRA.SZ[1:0] = 01b, and bit[1] and bit[0] must be 0 when MRA.SZ[1:0] = 10b, and bit[2], bit[1], and bit[0] must be 0 when MRA.SZ[1:0] = 11b.

18.2.7 CRA : DTC Transfer Count Register A

Base address: DTCVBR_SEC
DTCVBR

Offset address: 0x0E + 0x4 × Vector number
(Inaccessible directly from the CPU. See [section 18.3.1. Allocating Transfer Information and DTC Vector Table](#).)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
7:0	CRAL	Transfer Counter A Lower Register Specify the transfer count.	—
15:8	CRAH	Transfer Counter A Upper Register Specify the transfer count.	—

Note: The function depends on the transfer mode.

Note: Set CRAH and CRAL to the same value in repeat transfer mode and block transfer mode.

The CRA register consists of 16 bits. CRAL is the lower 8 bits and CRAH is the upper 8 bits. CRA is used in normal mode. CRAL and CRAH are used in repeat transfer mode and block transfer mode.

The CRA register cannot be accessed directly from the CPU. However, the CPU can access the SRAM area (transfer information (n) start address + 0x0E) and DTC transfers it automatically to and from the CRA register. See [section 18.3.1. Allocating Transfer Information and DTC Vector Table](#).

(1) Normal transfer mode (MRA.MD[1:0] = 00b)

In normal transfer mode, the CRA register functions as a 16-bit transfer counter. The transfer count is 1, 65535, and 65536 when the set value is 0x0001, 0xFFFF, and 0x0000, respectively. The CRA value is decremented (-1) on each data transfer.

(2) Repeat transfer mode (MRA.MD[1:0] = 01b)

In repeat transfer mode, the CRAH register holds the transfer count and the CRAL register functions as an 8-bit transfer counter. The transfer count is 1, 255, and 256 when the set value is 0x01, 0xFF, and 0x00, respectively. The CRAL value is decremented (-1) on each data transfer. When it reaches 0x00, the CRAH value is transferred to CRAL.

(3) Block transfer mode (MRA.MD[1:0] = 10b)

In block transfer mode, the CRAH register holds the block size and the CRAL register functions as an 8-bit block size counter. The transfer count is 1, 255, and 256 when the set value is 0x01, 0xFF, and 0x00, respectively. The CRAL value is decremented (-1) on each data transfer. When it reaches 0x00, the CRAH value is transferred to CRAL.

18.2.8 CRB : DTC Transfer Count Register B

Base address: DTCVBR_SEC
DTCVBR

Offset address: $0x0C + 0x4 \times \text{Vector number}$
(Inaccessible directly from the CPU. See [section 18.3.1. Allocating Transfer Information and DTC Vector Table.](#))

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

The CRB sets the block transfer count for block transfer mode. The transfer count is 1, 65535, and 65536 when the set value is 0x0001, 0xFFFF, and 0x0000, respectively. The CRB value is decremented (-1) when the final data of a single block size is transferred. When normal transfer mode or repeat transfer mode is selected, this register is not used, and the set value is ignored.

The CRB cannot be accessed directly from the CPU. However, the CPU can access the SRAM area (transfer information (n) start address + 0x0C) and DTC transfers it automatically to and from the CRB register. See [section 18.3.1. Allocating Transfer Information and DTC Vector Table.](#)

18.2.9 DTCCR : DTC Control Register

Base address: DTC0_NS = 0x5000_AC00

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RRS	—	—	—	—
Value after reset:	0	0	0	0	1	0	0	0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	—	This bit is read as 1. The write value should be 1.	R/W
4	RRS	DTC Transfer Information Read Skip Enable for Non-secure 0: Transfer information read is not skipped. 1: Transfer information read is skipped when vector numbers match	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE7, P-TYPE2

RRS bit (DTC Transfer Information Read Skip Enable for Non-secure)

The RRS bit enables skipping of transfer information reads when vector numbers match. The DTC vector number is compared with the vector number in the previous activation process. When these vector numbers match and the RRS bit is set to 1, DTC data transfer is performed without reading the transfer information. However, when the previous transfer is a chain transfer, the transfer information is read regardless of the RRS bit.

When the transfer counter (CRA register) becomes 0 during the previous normal transfer and when the transfer counter (CRB register) becomes 0 during the previous block transfer, the transfer information is read regardless of the RRS bit value.

Set the RRS bit to 0 when there is at least one transfer with the MRA.WBDIS bit set to 1.

If you want to set the MRC.DISPE bit to 1, you need to set the MRA.WBDIS bit to 1.

For information on the function of the RRS bit and how to set it, see [section 18.4.1. Transfer Information Read Skip Function.](#)

18.2.10 DTCCR_SEC : DTC Control Register for Secure Region

Base address: DTC0 = 0x4000_AC00

Offset address: 0x10

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RRS	—	—	—	—
Value after reset:	0	0	0	0	1	0	0	0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	—	This bit is read as 1. The write value should be 1.	R/W
4	RRS	DTC Transfer Information Read Skip Enable for Secure 0: Transfer information read is not skipped. 1: Transfer information read is skipped when vector numbers match.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE6, P-TYPE2

RRS bit (DTC Transfer Information Read Skip Enable for Secure)

The RRS bit enables skipping of transfer information reads when vector numbers match. The DTC vector number is compared with the vector number in the previous activation process. When these vector numbers match and the RRS bit is set to 1, DTC data transfer is performed without reading the transfer information. However, when the previous transfer is a chain transfer, the transfer information is read regardless of the RRS bit.

When the transfer counter (CRA register) becomes 0 during the previous normal transfer and when the transfer counter (CRB register) becomes 0 during the previous block transfer, the transfer information is read regardless of the RRS bit value.

18.2.11 DTCVBR : DTC Vector Base Register

Base address: DTC0_NS = 0x5000_AC00

Offset address: 0x04

Bit position:	31	0
Bit field:		
Value after reset:	0 0	

Bit	Symbol	Function	R/W
31:0	n/a	DTC Vector Base Address for Non-secure region Set the DTC vector base address for non-secure region. The lower 10 bits should be 0.	R/W

Note: S-TYPE7, P-TYPE2

The DTCVBR sets the base address for calculating the DTC vector table address, which can be set in the range of 0x0000_0000 to 0xFFFF_FFFF (4 GB) in 1 KB units.

For information on the function of the RRS bit and how to set it, see [section 18.4.1. Transfer Information Read Skip Function](#).

18.2.12 DTCVBR_SEC : DTC Vector Base Register for Secure Region

Base address: DTC0 = 0x4000_AC00

Offset address: 0x14

Bit position: 31

0

Bit field:

--

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	DTC Vector Base Address for Secure region Set the DTC vector base address for secure region. The lower 10 bits should be 0.	R/W

Note: S-TYPE6, P-TYPE2

The DTCVBR_SEC sets the base address for calculating the DTC vector table address, which can be set in the range of 0x0000_0000 to 0xFFFF_FFFF (4 GB) in 1-KB units.

18.2.13 DTCST : DTC Module Start Register

Base address: DTC0 = 0x4000_AC00
DTC0_NS = 0x5000_AC00

Offset address: 0x0C

Bit position: 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	—	—	DTCST
---	---	---	---	---	---	---	-------

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DTCST	DTC Module Start 0: DTC module stopped 1: DTC module started	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE2

DTCST bit (DTC Module Start)

Set the DTCST bit to 1 to enable the DTC to accept transfer requests. When this bit is set to 0, transfer requests are no longer accepted. If this bit is set to 0 during a data transfer, the accepted transfer request is active until processing completes.

DTCST must be set to 0 before transitioning to one of the following state or mode:

- Module-stop state
- Software Standby mode
- Deep Software Standby mode

For details on these transitions, see [section 18.10. Low Power Consumption Function](#) and [section 11, Low Power Mode](#).

18.2.14 DTCSTS : DTC Status Register

Base address: DTC0 = 0x4000_AC00
DTC0_NS = 0x5000_AC00

Offset address: 0x0E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ACT	—	—	—	—	—	—	—	VECN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	VECN[7:0]	DTC-Activating Vector Number Monitoring These bits indicate the vector number for the activation source when a DTC transfer is in progress. The value is only valid if a DTC transfer is in progress (ACT flag is 1).	R
14:8	—	These bits are read as 0.	R
15	ACT	DTC Active Flag 0: DTC transfer operation is not in progress. 1: DTC transfer operation is in progress.	R

Note: S-TYPE3, P-TYPE2

VECN[7:0] bits (DTC-Activating Vector Number Monitoring)

While transfer by the DTC is in progress, the VECN[7:0] bits indicate the vector number associated with the activation source for the transfer. The value read from the VECN[7:0] bits is valid if the ACT flag is 1, indicating a DTC transfer in progress, and invalid if the ACT flag is 0, indicating no DTC transfer is in progress.

ACT flag (DTC Active Flag)

The ACT flag indicates the state of the DTC transfer operation.

[Setting condition]

- When the DTC is activated by a transfer request.

[Clearing condition]

- When transfer by the DTC, in response to a transfer request, is complete.

18.2.15 DTCDISP : DTC Address Displacement Register

Base address: DTC0 = 0x4000_AC00
DTC0_NS = 0x5000_AC00

Offset address: 0x18

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DTCDISP[31:16]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DTCDISP[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	DTCDISP[31:0]	DTC Address Displacement Displacement value to be added to the DTC forwarding source address.	R/W

Note: S-TYPE-7, P-TYPE-2

The DTCDISP register specifies the displacement value to be added to the DTC forwarding source address. If the MRC.DISPE bit is 1, the value of SAR + DTCDISP is used as the transfer source address.

18.2.16 DTEVR : DTC Error Vector Register

Base address: DTC0 = 0x4000_AC00
DTC0_NS = 0x5000_AC00

Offset address: 0x20

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DTESTA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	DTEVSAM	DTEV[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	DTEV[7:0]	DTC Error Vector Number These bits represent error vector of the DTC.	R
8	DTEVSAM	DTC Error Vector Number SA Monitor Indicates the SA of vector number causing the error. 0: Secure vector number 1: Non-secure vector number	R
15:9	—	These bits are read as 0. The write value should be 0.	R
16	DTESTA	DTC Error Status Flag 0: No DTC transfer error occurred 1: DTC transfer error occurred	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R

Note: P-TYPE2 See below for access by security attribution.

Note: Writing to DTESTA depends on the value of DTEVSAM

When reading this register, it can be accessed from both secure and non-secure access.

When writing this register, it depends on DTEVR.DTEVSAM.

- When DTEVSAM = 1, it can be accessed from Secure and Non-secure access.
- When DTEVSAM = 0, it can be accessed from Secure. An error is returned when write access is performed in a Non-secure access.

This register is cleared by a reset caused by a transfer error. Please select interrupt DMAM_TRANSERR (m = 0, 1) in BUS.BUSOAD and TZF.MSAOAD when you want to debug the program.

DTEV[7:0] bit (DTC Error Vector Number)

When a transfer error due to DTC transfer occurs, it stores the channel of DTC that was violated.

[Setting condition]

- When the DTC transfer error occurs and DTESTA = 0.

[Clearing condition]

- When 1 is written to DTEVR.DTESTA.

DTEVSAM bit (DTC Error Vector Number SA Monitor)

When a transfer error due to DTC transfer occurs, it indicates the SA of the violating DTC vector number.

[Setting condition]

- When the DTC transfer error occurs and DTESTA = 0.

[Clearing condition].

- When 1 is written to DTEVR.DTESTA.

DTESTA bit (DTC Error Status Flag)

Indicates whether or not a DTC transfer error occurred.

DTEV, DTEVSAM, DTESTA are cleared by writing 1 to DTESTA.

Writing 0 to DTESTA is ignored.

[Setting condition]

- When the DTC transfer error occurs.

[Clearing condition]

- When 1 is written to DTEVR.DTESTA.

18.3 Activation Sources

The DTC is activated by an interrupt request. Setting the ICU.IELSRn.DTCE bit to 1 enables activation of the DTC by the associated interrupt. The selector output n number set in ICU.IELSRn is defined as the interrupt vector number. For an enabled interrupt, the specific DTC interrupt source associated with each interrupt vector number n is selected in ICU.IELSRn.IELS[9:0] ($n = 0$ to 95), as listed in [section 14.3.2. Event Number](#). For activation by software, see [section 20.2.2. ELSEGRn : Event Link Software Event Generation Register \$n\$ \(\$n = 0\$ to 3\)](#).

The interrupt vector number is equivalent to the DTC vector table number. After the DTC accepted an activation request, it does not accept another activation request until the transfer for that single request is complete, regardless of the priority of the requests. When multiple activation requests are generated during a DTC transfer, the highest priority request is accepted on completion of the transfer. When multiple activation requests are generated while the DTC Module Start bit (DTCST.DTCST) is 0, the DTC accepts the highest priority request when DTCST.DTCST is subsequently set to 1. The smaller interrupt vector number has higher priority.

The DTC performs the following operations at the start of a single data transfer or for a chain transfer, after the last of the consecutive transfers:

- On completion of a specified round of data transfer, the ICU.IELSRn.DTCE bit is set to 0, and an interrupt request is sent to the CPU.
- If the MRB.DISEL bit is 1, an interrupt request is sent to the CPU on completion of a data transfer.
- For other transfers, the ICU.IELSRn.IR flag of the activation source is set to 0 at the start of the data transfer.

18.3.1 Allocating Transfer Information and DTC Vector Table

The DTC reads the start address of the transfer information associated with each activation source from the vector table and reads the transfer information starting at that address.

The DTC has a vector table for secure DTC activation requests and a vector table for non-secure DTC activation requests. Place the vector table of the interrupt vector number of SA = 1 in DTCVBR which is the Non-secure side. Place the vector table of interrupt number SA = 0 in DTCVBR_SEC which is the Secure side.

The vector table must be located so that the lower 10 bits of the base address (start address) are 0. Use the DTC Vector Base Register (DTCVBR) to set the base address of the DTC vector table. Transfer information is allocated in the SRAM area. However, if the MRA.WBDIS bit is set to 1 (no write back), it can also be placed in the ROM area. The start address of the transfer information n with vector number n must be $4n$ added to the base address in the vector table. Each transfer information start address must be a multiple of 16.

When setting the DTC transfer attribution for the privilege access or unprivileged access, allocating the vector table and transfer information as follows. The vector table should be located in the privileged area. Transfer information with the privileged attribution is located in the privileged area. Transfer information with the unprivileged attribution is located in any privileged area. Privileged area is set by MPMU (see [section 16, Memory Protection Unit \(MPU\)](#)). Privileged and non-privileged access settings are located at bit 0 of the address corresponding to each vector number in the vector table.

[Figure 18.2](#) shows the relationship between the DTC vector table and transfer information.

[Figure 18.3](#) shows the bit arrangement in the vector table.

Figure 18.4 shows the allocation of transfer information in the SRAM area.

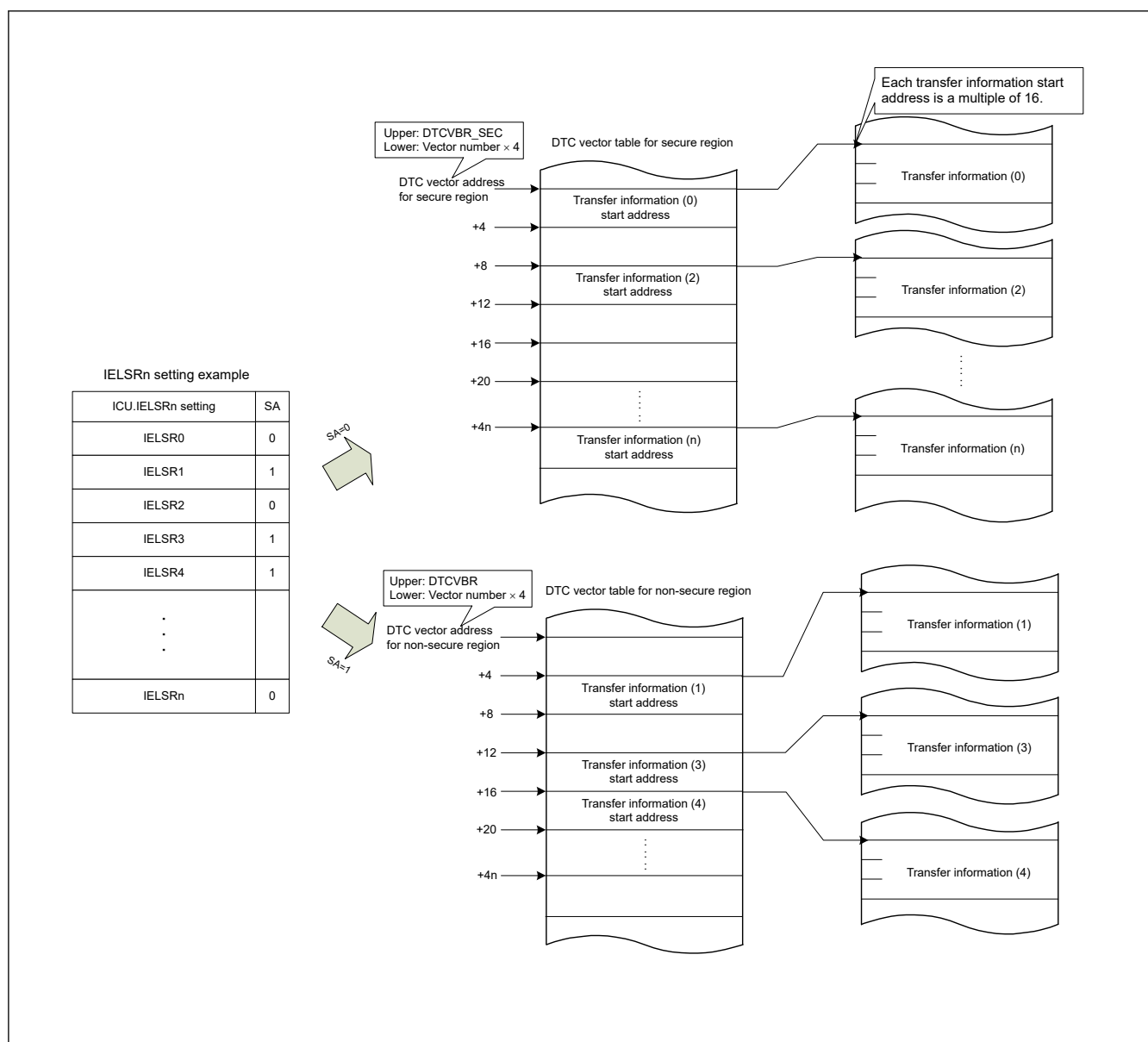


Figure 18.2 DTC vector table and transfer information

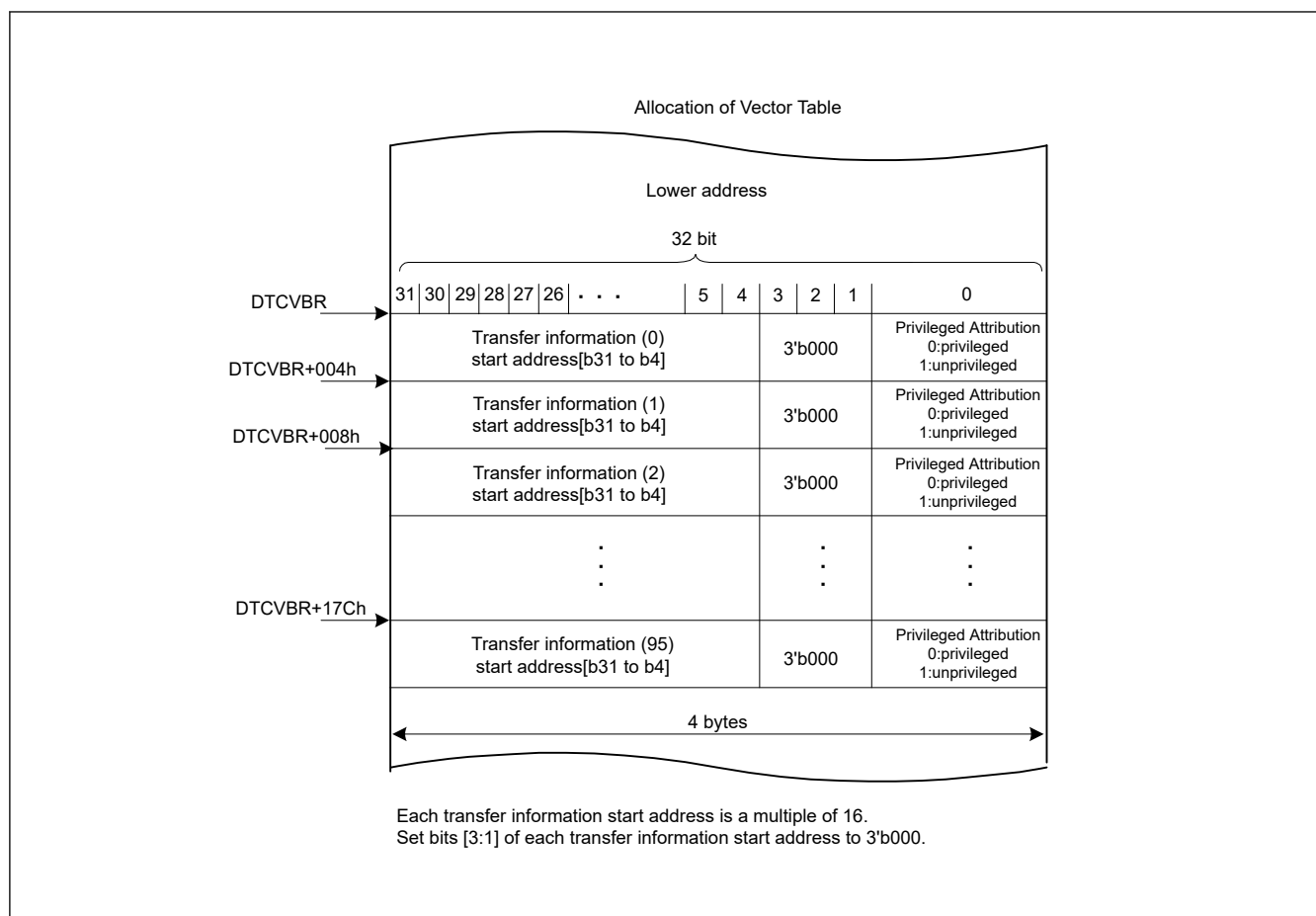


Figure 18.3 Bit arrangement in the vector table

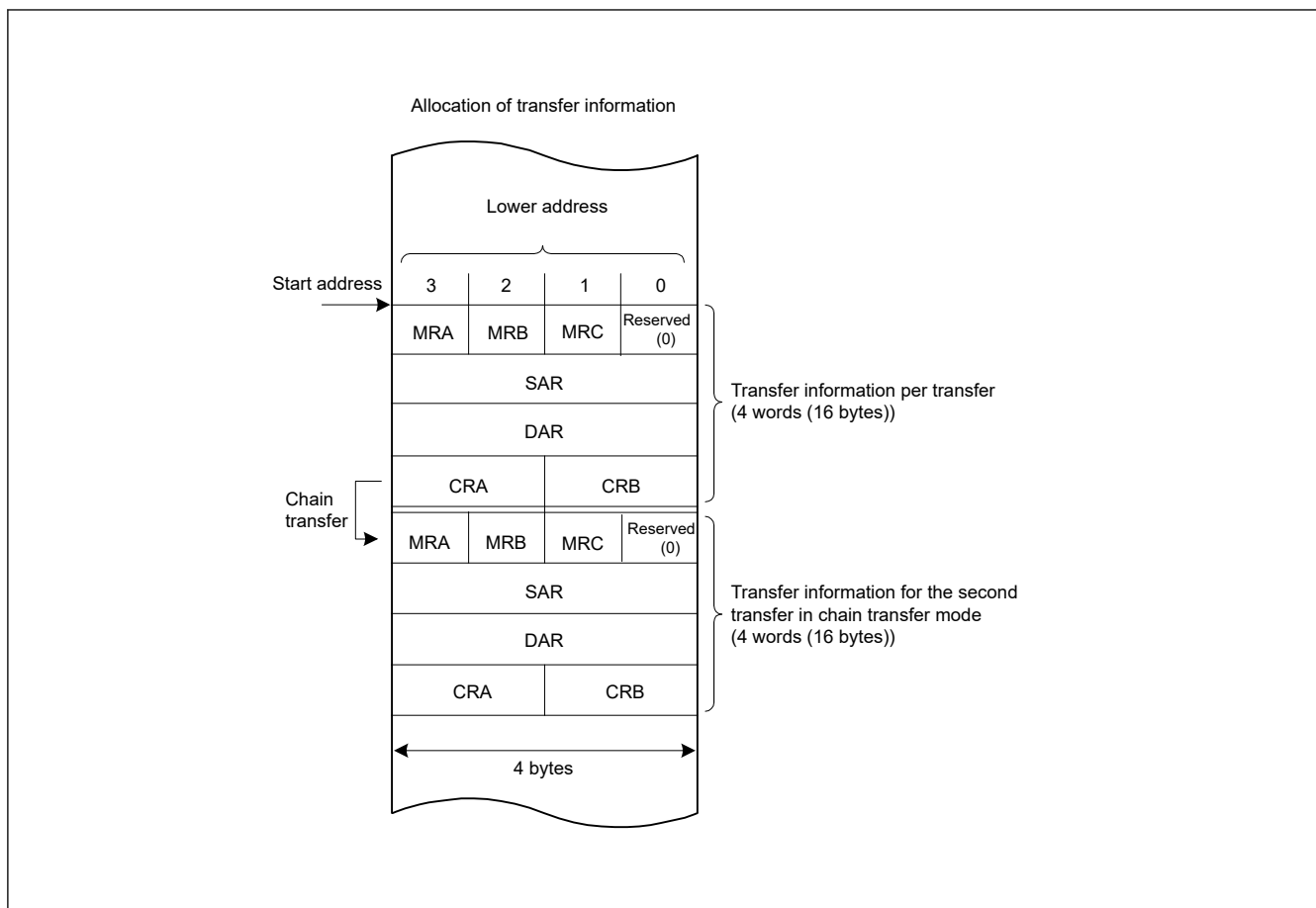


Figure 18.4 Allocation of transfer information in the SRAM area

18.4 Operation

The DTC transfers data according to the transfer information. Storage of the transfer information in the SRAM area is required before a DTC operation. When the DTC is activated, it reads the DTC vector associated with the vector number. The DTC reads the transfer information from the transfer information store address referenced by the DTC vector and transfers the data. After the data transfer, the DTC writes back the transfer information. Storing the transfer information in the SRAM area allows data transfer of any number of channels.

The transfer modes include:

- Normal transfer mode
- Repeat transfer mode
- Block transfer mode.

The DTC specifies a transfer source address in the SAR register and a transfer destination address in the DAR register. The values of these registers are incremented, decremented, or address-fixed independently after the data transfer.

Table 18.2 describes the DTC transfer modes.

Table 18.2 DTC transfer modes (1 of 2)

Transfer mode	Data size transferred on single transfer request	Increment or decrement of memory address	Settable transfer count
Normal transfer mode	1 byte (8 bits), 1 halfword (16 bits), 1 word (32 bits), or 1 doubleword (64 bits).	Incremented or decremented by 1, 2, 4, or 8 or address-fixed	1 to 65536
Repeat transfer mode*1	1 byte (8 bits), 1 halfword (16 bits), 1 word (32 bits), or 1 doubleword (64 bits).	Incremented or decremented by 1, 2, 4, or 8 or address-fixed	1 to 256*3

Table 18.2 DTC transfer modes (2 of 2)

Transfer mode	Data size transferred on single transfer request	Increment or decrement of memory address	Settable transfer count
Block transfer mode ^{*2}	Block size specified in CRAH (1 to 256 bytes, 1 to 256 halfwords (2 to 512 bytes), 1 to 256 words (4 to 1024 bytes), or 1 to 256 doublewords (8 to 2048 bytes)).	Incremented or decremented by 1, 2, 4, or 8 or address-fixed	1 to 65536

Note 1. Set the transfer source or transfer destination as the repeat area.

Note 2. Set the transfer source or transfer destination as the block area.

Note 3. After a data transfer of the specified count, the initial state is restored and operation restarts.

Setting the MRB.CHNE bit to 1 allows multiple transfers or chain transfer on a single activation source. It also enables a chain transfer when the specified data transfer is complete.

Figure 18.5 shows the operation flow of the DTC. Table 18.3 lists the chain transfer conditions. The combination of control information for the second and subsequent transfers are omitted in this table.

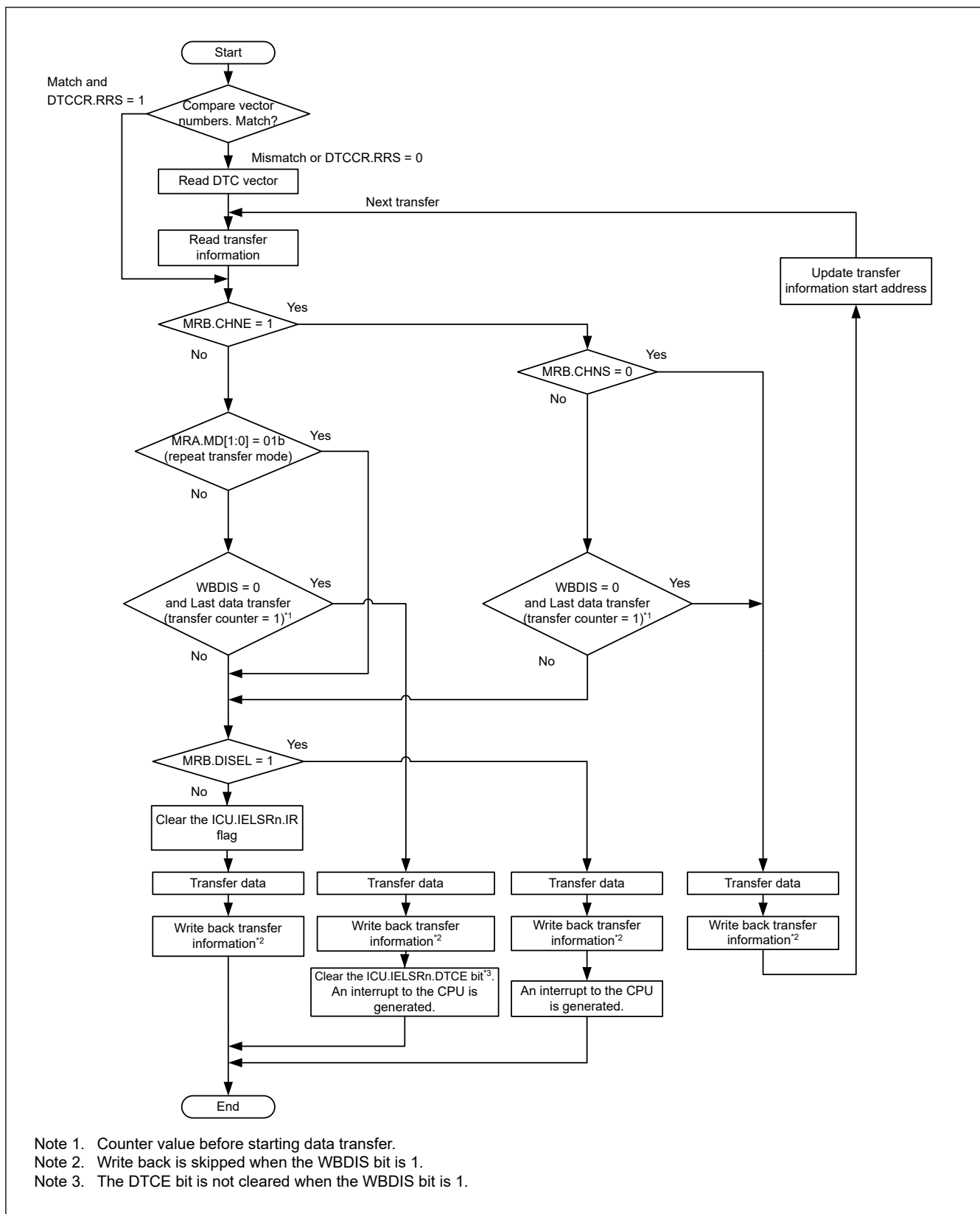


Figure 18.5 DTC operation flow

Table 18.3 Chain transfer conditions

First transfer				Second transfer ^{*3}				DTC transfer
CHNE bit	CHNS bit	DISEL bit	Transfer counter ^{*1 *2}	CHNE bit	CHNS bit	DISEL bit	Transfer counter ^{*1 *2}	
0	—	0	Other than (1 → 0)	—	—	—	—	Ends after the first transfer
0	—	0	(1 → 0)	—	—	—	—	Ends after the first transfer with an interrupt request to the CPU
0	—	1	—	—	—	—	—	
1	0	—	—	0	—	0	Other than (1 → 0)	Ends after the second transfer
				0	—	0	(1 → 0)	Ends after the second transfer with an interrupt request to the CPU
				0	—	1	—	
1	1	0	Other than (1 → *)	—	—	—	—	Ends after the first transfer
1	1	—	(1 → *)	0	—	0	Other than (1 → 0)	Ends after the second transfer
				0	—	0	(1 → 0)	Ends after the second transfer with an interrupt request to the CPU
				0	—	1	—	
1	1	1	Other than (1 → *)	—	—	—	—	Ends after the first transfer with an interrupt request to the CPU

Note 1. The transfer counter used depends on the transfer modes as follows:

Normal transfer mode — CRA register
Repeat transfer mode — CRAL register
Block transfer mode — CRB register

Note 2. On completion of a data transfer, the counters operate as follows:

1 → 0 in normal and block transfer modes
1 → CRAH in repeat transfer mode
(1 → *) in the table indicates both of these two operations, depending on the mode.

Note 3. Chain transfer can be selected for the second or subsequent transfers. The conditions for the combination of the second transfer and CHNE = 1 is omitted.

18.4.1 Transfer Information Read Skip Function

Reading of vector addresses and transfer information can be skipped by setting the DTCCR.RRS bit. When a DTC activation request is generated, the current DTC vector number is compared with the DTC vector number in the previous activation process. When these vector numbers match and the RRS bit is set to 1, the DTC data transfer is performed without reading the vector address and transfer information. However, when the previous transfer is a chain transfer, the vector address and transfer information are read. Additionally, when the transfer counter (CRA register) becomes 0 during the previous normal transfer, and when the transfer counter (CRB register) becomes 0 during the previous block transfer, transfer information is read regardless of the RRS bit. [Figure 18.13](#) shows an example when reading the transfer information is skipped.

To update the vector table and transfer information, set the RRS bit to 0, update the vector table and transfer information, then set the RRS bit to 1. The stored vector number is discarded by setting the RRS bit to 0. The updated DTC vector table and transfer information are read in the next activation process.

If there is at least one transfer information with the MRA.WBDIS bit set to 1, the transfer information read skip function cannot be used. Set the DTCCR.RRS bit to 0 (do not perform read skip). When used in secure state, access DTCCR_SEC.RRS instead of DTCCR.RRS.

18.4.2 Transfer Information Write-Back Skip Function

When the MRA.SM[1:0] bits or the MRB.DM[1:0] bits are set to address fixed, a part of the transfer information is not written back. [Table 18.4](#) lists the transfer information write-back skip conditions and the associated registers. The CRA and CRB registers are written back, and the write-back of the MRA and MRB registers is skipped.

When the MRA.WBDIS bit is 1, all transfer information (SAR, DAR, CRA, CRB) is not written back. Since the transfer information in memory is not updated, DTC data transfer can be performed without copying the transfer information from ROM to RAM. Also, by omitting writeback, the time required for post-processing of data transfer can be shortened.

Table 18.4 Transfer information write-back skip conditions and applicable registers

MRA.SM[1:0] bits		MRB.DM[1:0] bits		SAR register	DAR register
b3	b2	b3	b2		
0	0	0	0	Skip	Skip
0	0	0	1		
0	1	0	0		
0	1	0	1		
0	0	1	0	Skip	Write-back
0	0	1	1		
0	1	1	0		
0	1	1	1		
1	0	0	0	Write-back	Skip
1	0	0	1		
1	1	0	0		
1	1	0	1		
1	0	1	0	Write-back	Write-back
1	0	1	1		
1	1	1	0		
1	1	1	1		

18.4.3 Normal Transfer Mode

The normal transfer mode allows a 1-byte (8 bits), 1-halfword (16 bits), 1-word (32 bits), or 1-doubleword (64 bits) data transfer on a single activation source. The transfer count can be set from 1 to 65536. Transfer source and destination addresses can be independently set to increment, decrement, or fixed. This mode enables an interrupt request to the CPU to be generated at the end of a specified-count transfer.

[Table 18.5](#) lists register functions in normal transfer mode, and [Figure 18.6](#) shows the memory map of normal transfer mode.

Table 18.5 Register functions in normal transfer mode

Register	Description	Value written back by writing transfer information*2
SAR	Transfer source address	Increment, decrement, or fixed*1
DAR	Transfer destination address	Increment, decrement, fixed*1
CRA	Transfer counter A	CRA - 1
CRB	Transfer counter B	Not updated

Note 1. Write-back operation is skipped in address-fixed mode.

Note 2. Write-back is skipped when MRA.WBDIS = 1.

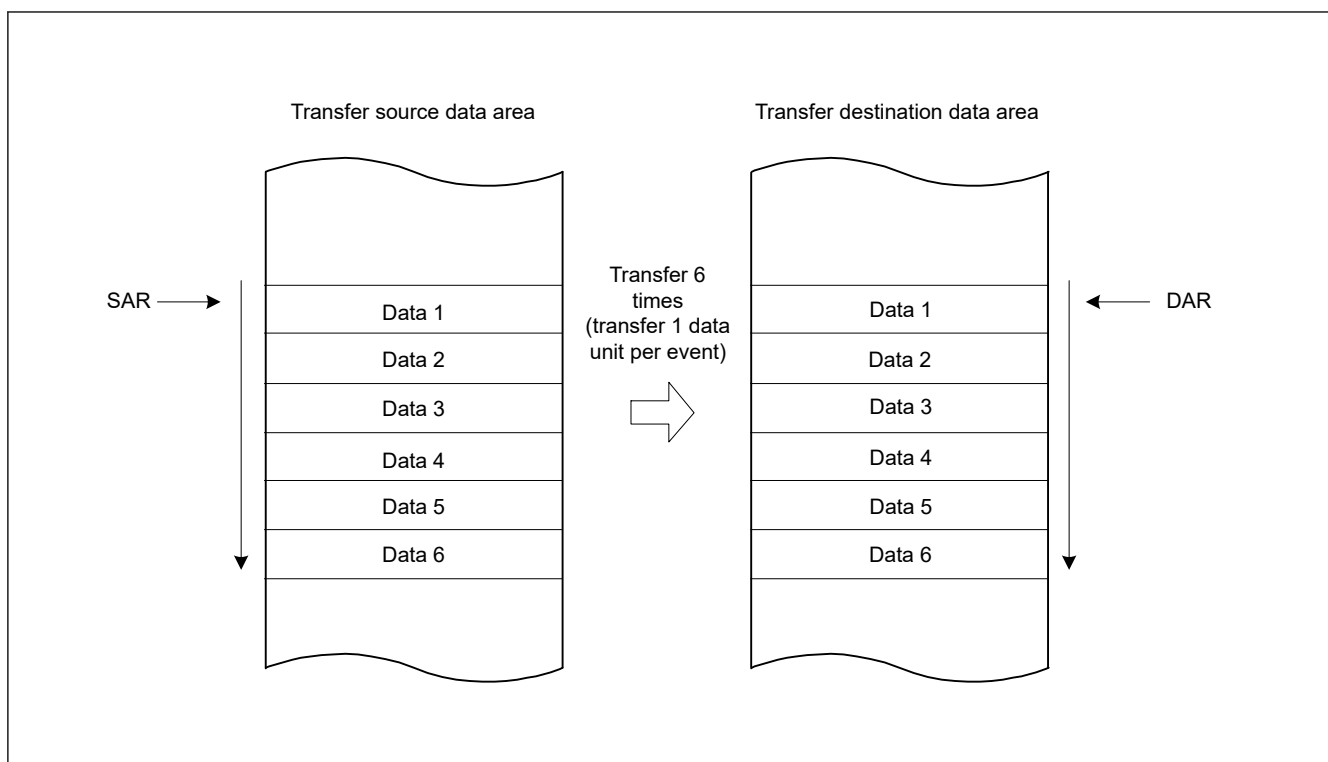


Figure 18.6 Memory map of normal transfer mode (MRA.SM[1:0] = 10b, MRB.DM[1:0] = 10b, CRA = 0x0006)

18.4.4 Repeat Transfer Mode

The repeat transfer mode allows a 1-byte (8 bits), 1-halfword (16 bits), 1-word (32 bits), or 1-doubleword (64 bits) data transfer on a single activation source. Transfer source or transfer destination for the repeat area must be specified in the MRB.DTS bit. The transfer count can be set from 1 to 256. When the specified transfer count is complete, the initial value of the address register specified in the repeat area is restored, the initial value of the transfer counter is restored, and transfer is repeated. The other address register is incremented or decremented continuously or remains unchanged.

When the transfer counter CRAL decrements to 0x00 in repeat transfer mode, the CRAL value is updated to the value set in the CRAH register. As a result, the transfer counter does not clear to 0x00, which disables interrupt requests to the CPU when the MRB.DISEL bit is set to 0. An interrupt request to the CPU is generated when the specified data transfer completes.

Table 18.6 lists the register functions in repeat transfer mode, and Figure 18.7 shows the memory map of repeat transfer mode.

Table 18.6 Register functions in repeat transfer mode

Register	Description	Value written back by writing transfer information*2	
		When CRAL is not 1	When CRAL is 1
SAR	Transfer source address	Increment, decrement, fixed*1	- When the MRB.DTS bit is 0 Increment, decrement, or fixed*1 - When the MRB.DTS bit is 1 SAR register initial value
DAR	Transfer destination address	Increment, decrement, or fixed*1	- When the MRB.DTS bit is 0 DAR register initial value - When the MRB.DTS bit is 1 Increment, decrement, or fixed*1
CRAH	Retains transfer counter	CRAH	CRAH
CRAL	Transfer counter A	CRAL - 1	CRAH
CRB	Transfer counter B	Not updated	Not updated

Note 1. Write-back is skipped in address-fixed mode.

Note 2. Write-back is skipped when MRA.WBDIS = 1.

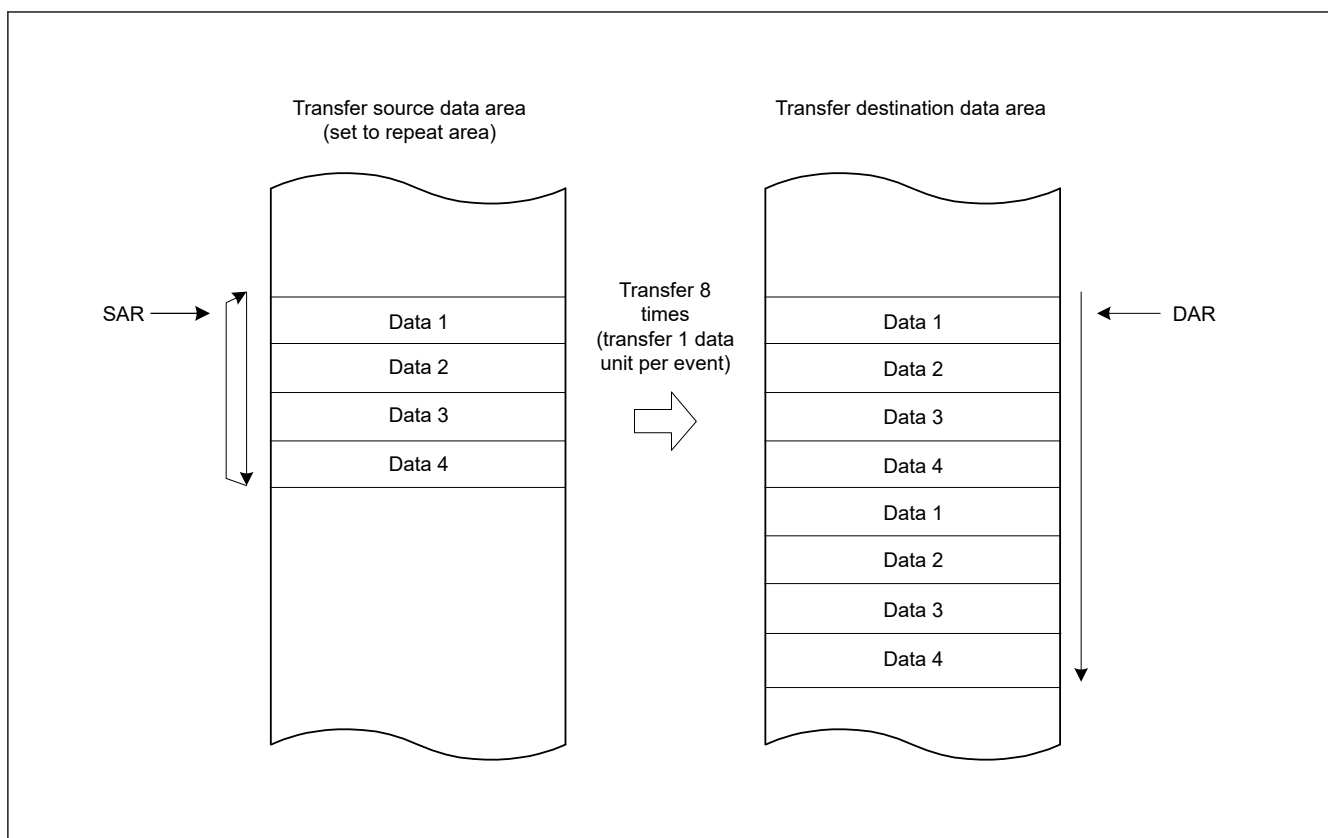


Figure 18.7 Memory map of repeat transfer mode when transfer source is a repeat area (MRA.SM[1:0] = 10b, MRB.DM[1:0] = 10b, CRAH = 0x04)

18.4.5 Block Transfer Mode

The block transfer mode allows single-block data transfer on a single activation source. Transfer source or transfer destination for the block area must be specified in the MRB.DTS bit. The block size can be set from 1 to 256 bytes, 1 to 256 halfwords (2 to 512 bytes), 1 to 256 words (4 to 1024 bytes), or 1 to 256 doublewords (8 to 2048 bytes). When transfer of the specified block completes, the initial values of the block size counter CRAL and the address register (the SAR register when the MRB.DTS = 1 or the DAR register when the DTS = 0) specified in the block area are restored. The other address register is incremented or decremented continuously or remains unchanged.

The transfer count (block count) can be set from 1 to 65536. This mode enables an interrupt request to the CPU to be generated at the end of the specified-count block transfer.

Table 18.7 lists the register functions in block transfer mode, and Figure 18.8 shows the memory map for block transfer mode.

Table 18.7 Register functions in block transfer mode

Register	Description	Value written back by writing transfer information ^{*2}
SAR	Transfer source address	- When MRB.DTS bit is 0 Increment, decrement, or fixed^{*1} - When MRB.DTS bit is 1 SAR register initial value.
DAR	Transfer destination address	- When MRB.DTS bit is 0 DAR register initial value - When MRB.DTS bit is 1 Increment, decrement, or fixed^{*1}.
CRAH	Holds block size	CRAH
CRAL	Block size counter	CRAH
CRB	Block transfer counter	CRB - 1

Note 1. Write-back is skipped in address-fixed mode.

Note 2. Write-back is skipped when MRA.WBDIS = 1.

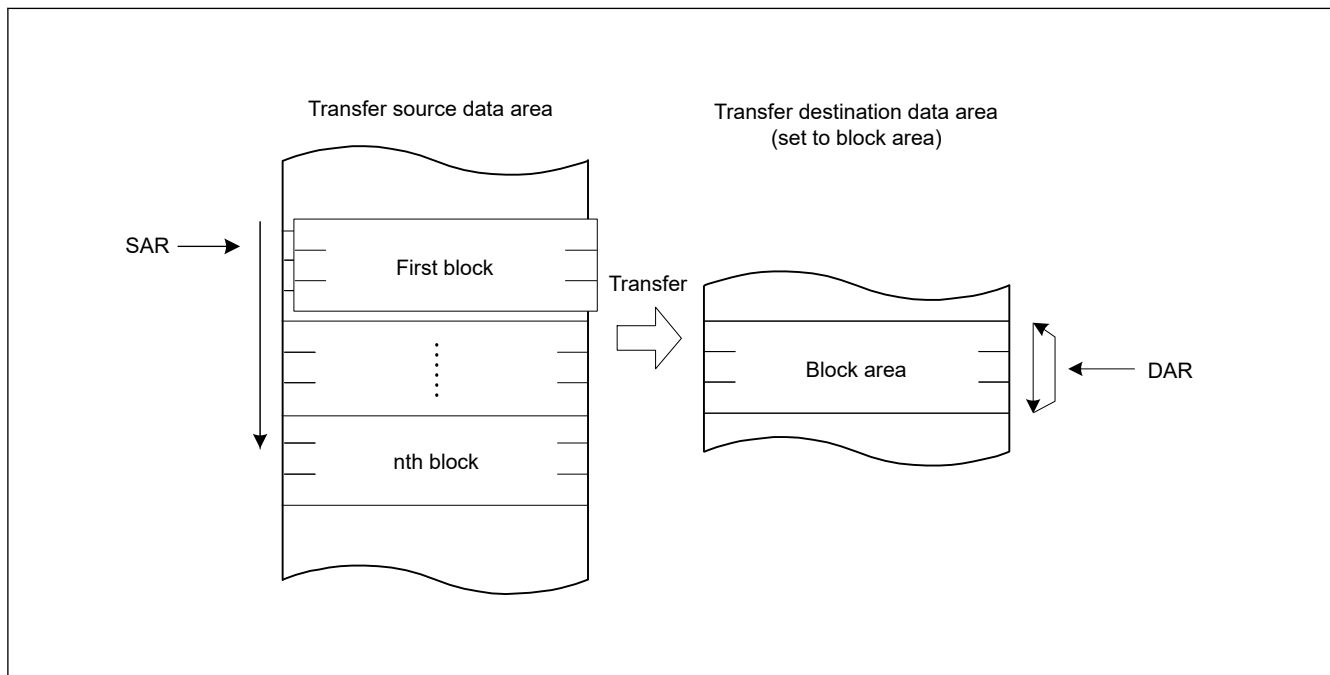


Figure 18.8 Memory map of block transfer mode

18.4.6 Chain Transfer

Setting the MRB.CHNE bit to 1 allows chain transfer to be performed continuously on a single activation source. If the MRB.CHNE is set to 1 and CHNS to 0, an interrupt request to the CPU is not generated on completion of the specified number of rounds of transfer or by setting the MRB.DISEL bit to 1. An interrupt request is sent to the CPU each time DTC data transfer is performed. Data transfer has no effect on the ICU.IELSRn.IR flag of the activation source.

The SAR, DAR, CRA, CRB, MRA, MRB, and MRC registers can be set independently of each other to define the data transfer. [Figure 18.9](#) shows a chain transfer operation.

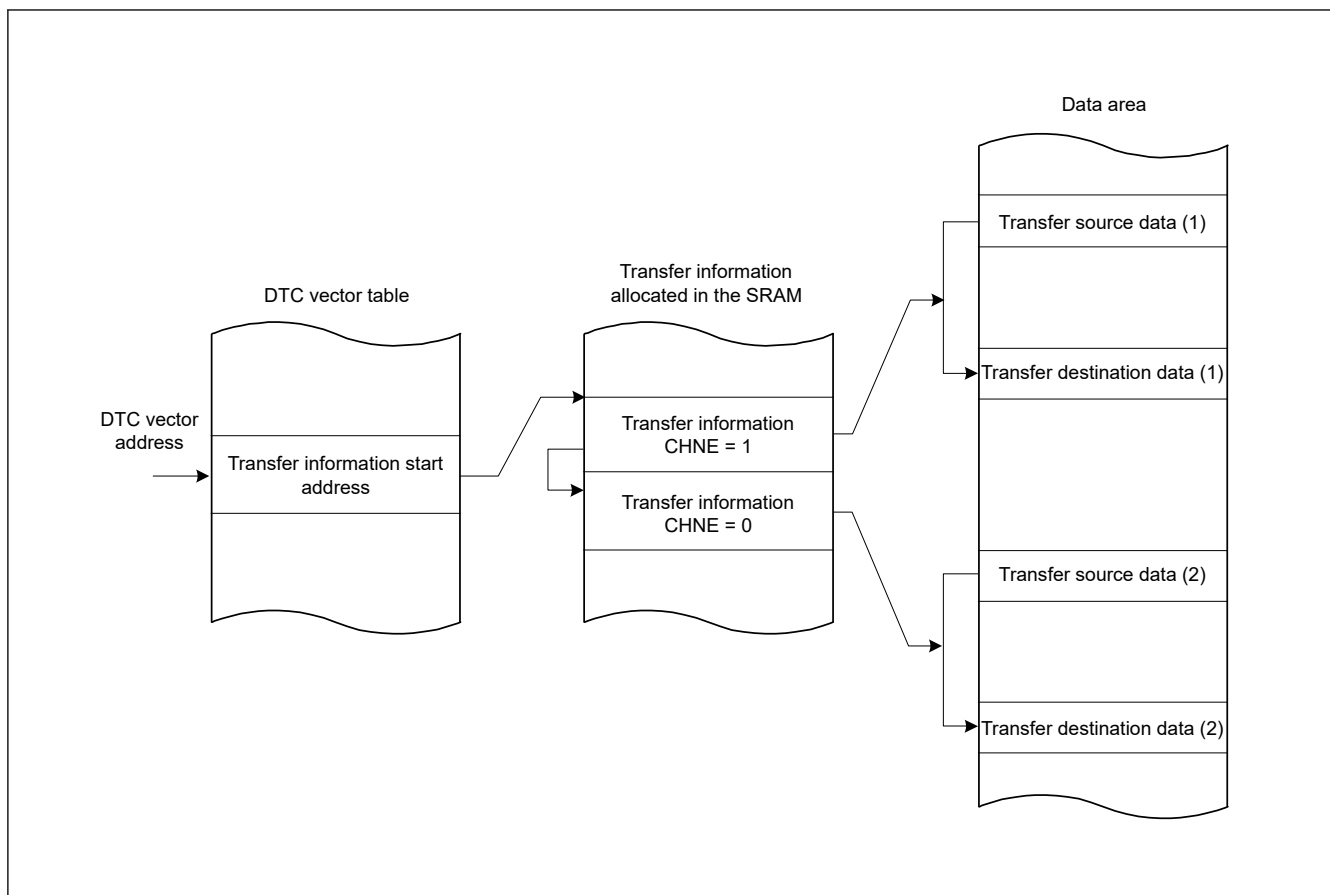


Figure 18.9 Chain transfer operation

Writing 1 to the MRB.CHNE and CHNS bits enables chain transfer to be performed only after completion of the specified data transfer. In repeat transfer mode, chain transfer is performed after completion of the specified data transfer. For details on chain transfer conditions, see [Table 18.3](#).

18.4.7 Operation Timing

[Figure 18.10](#) to [Figure 18.13](#) are timing diagrams that show the minimum number of execution cycles.

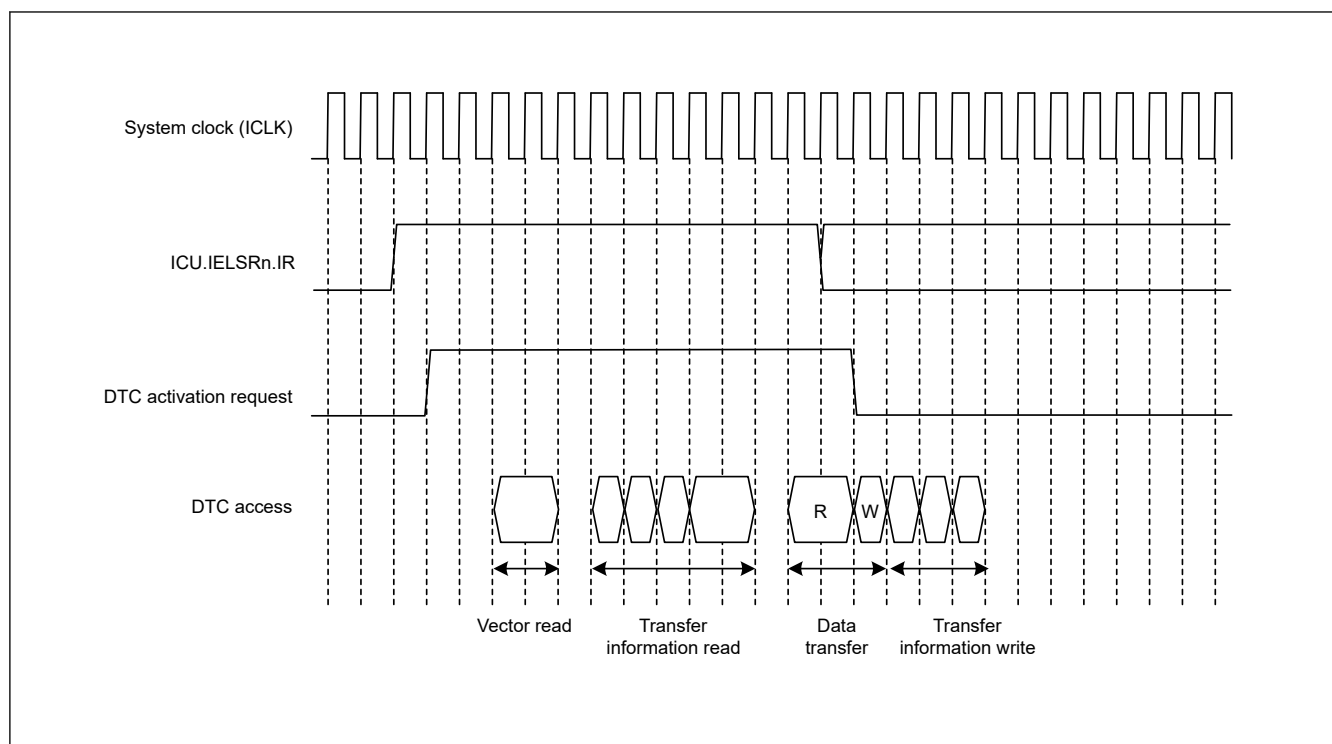


Figure 18.10 Example 1 of DTC operation timing in normal transfer and repeat transfer modes

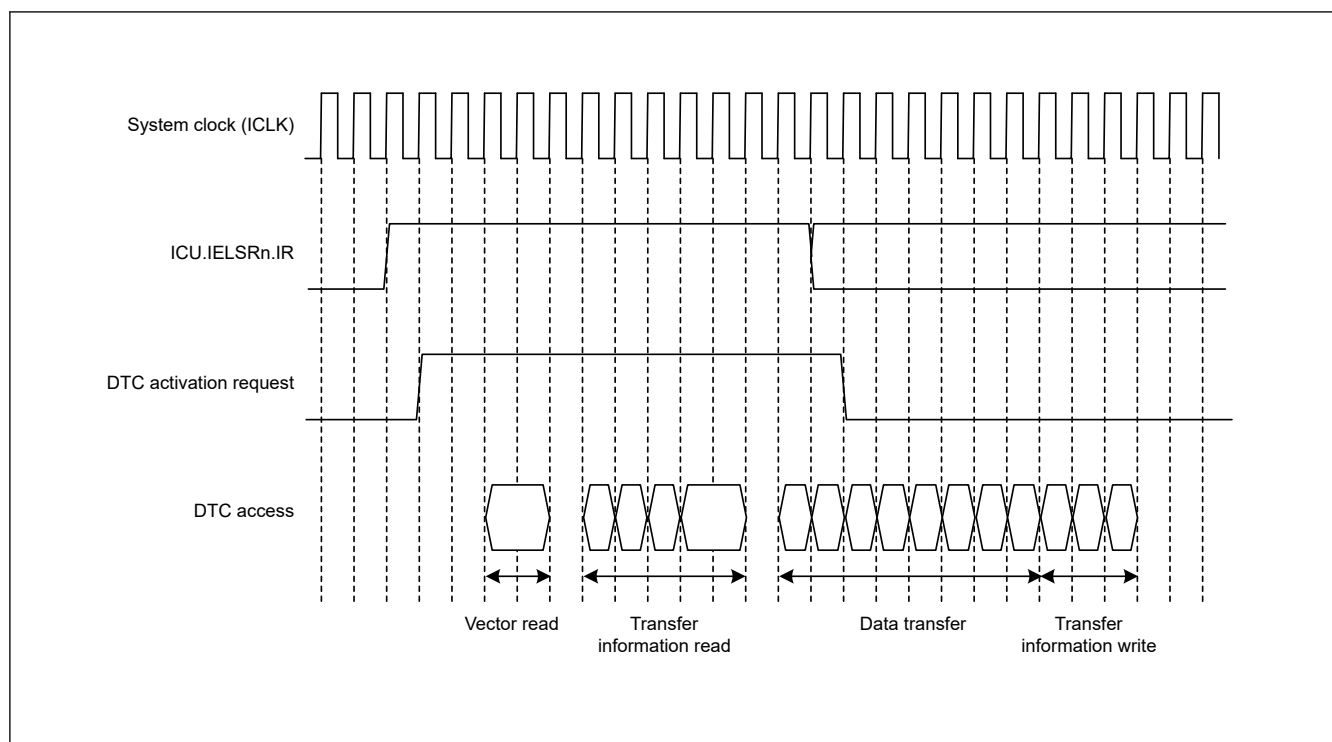


Figure 18.11 Example 2 of DTC operation timing in block transfer mode when the block size = 4

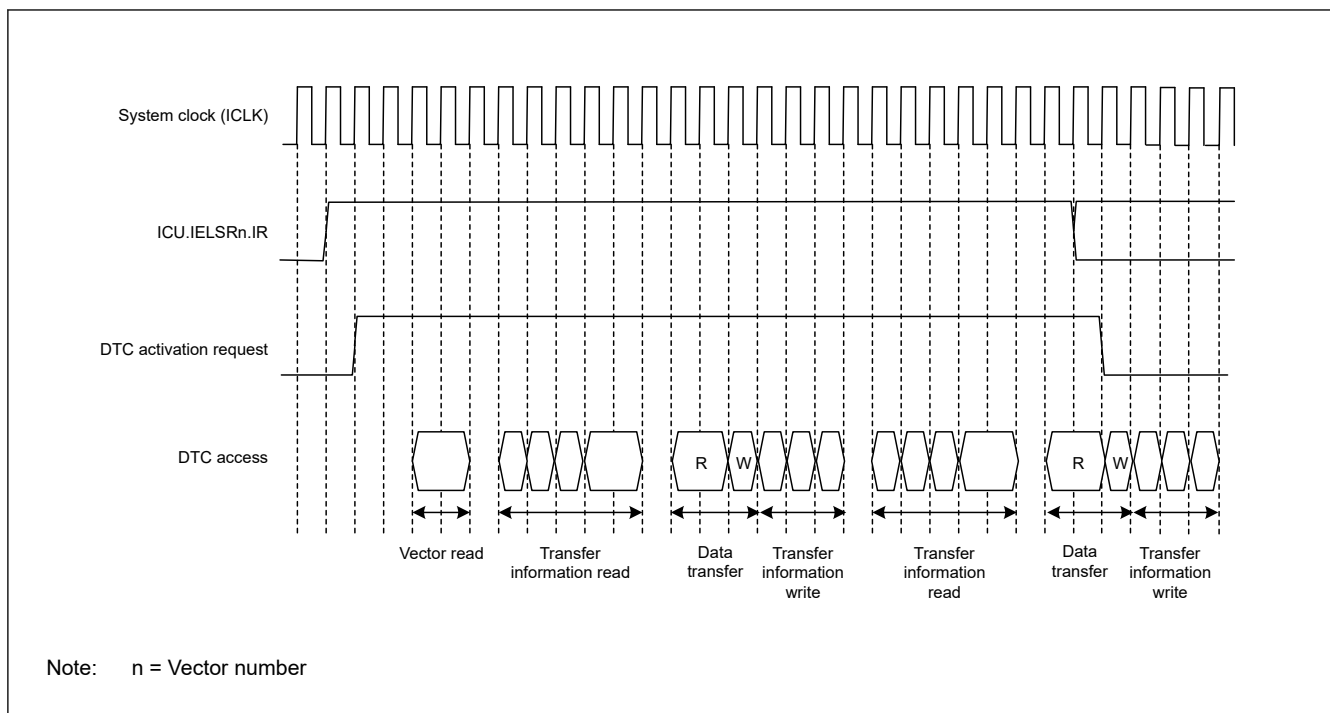


Figure 18.12 Example 3 of DTC operation timing for chain transfer

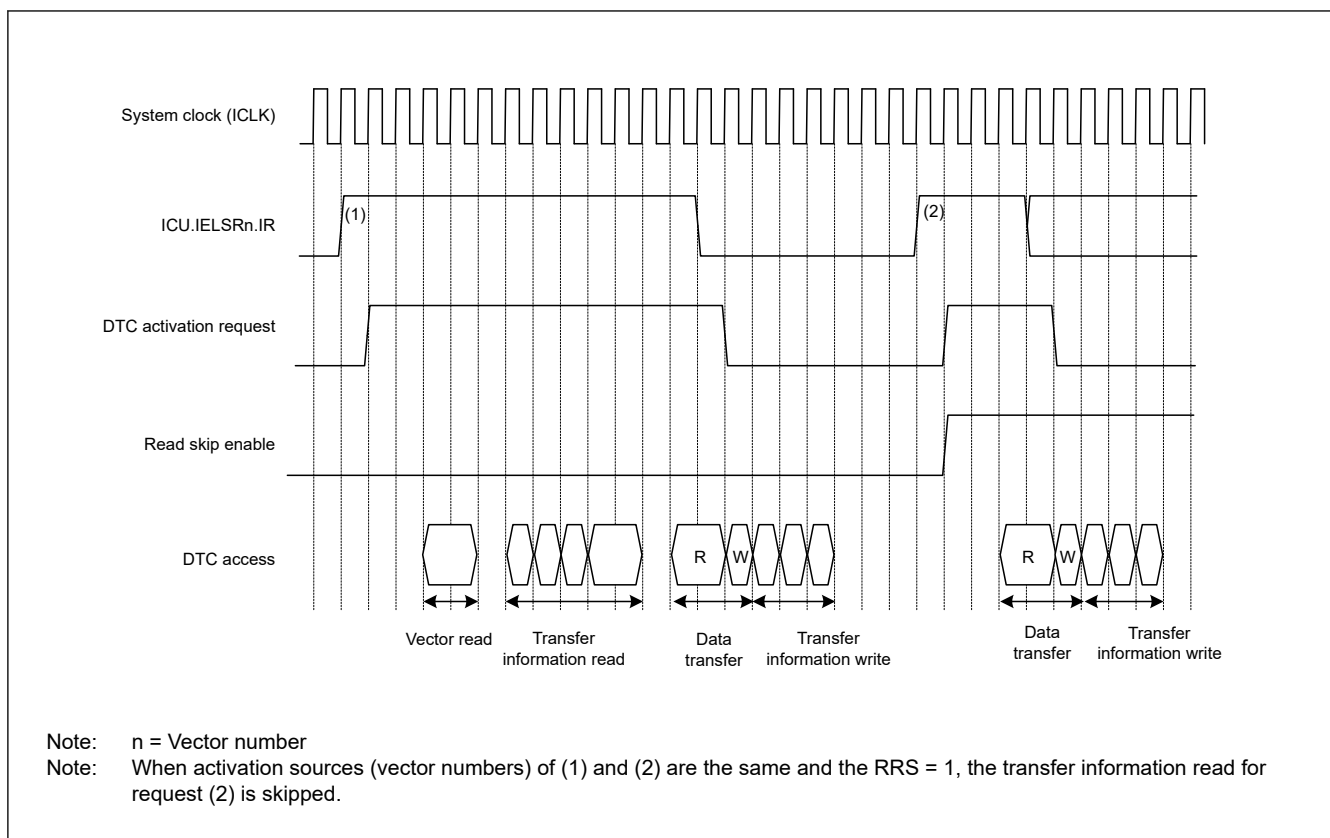


Figure 18.13 Example of operation when a transfer information read is skipped with the vector, transfer information, and transfer destination data on the SRAM, and the transfer source data on the peripheral module

18.4.8 Vector Security

The security attribute of transfer access of DTC vector n and security attribute of access to the IELSR n ($n = 0$ to 95) register of ICU are controlled by SAIELSR n bit of ICUSAR x ($x = G, H, I, J, K$ or L) registers in CPSCU. For details on the CPSCU.ICUSAR x registers, see [section 14, Interrupt Controller Unit \(ICU\)](#).

When the CPSCU.ICUSAR x .SAIELSR n bit is 0

- IELSR n register have security attribute
- The vector number of DTC activated by the interrupt request set in IELS behaves as a secure master.

When the CPSCU.ICUSAR x .SAIELSR n bit is 1

- IELSR n register have non-security attribute
- The vector number of DTC activated by the interrupt request set in IELS behaves as a non-secure master.

For areas accessible to secure or non-secure masters, See [section 15, Buses](#), and [section 52, Security Features](#).

18.4.9 Vector Privilege

The privileged attribute of transfer access of DTC vector n is controlled bit0 in vector table.

If bit0 corresponding to each vector number in the vector table is set to 0, DTC transfer is privileged access.

If bit0 is set to 1, DTC transfers are unprivileged access.

Refer to for vector table and transfer information.

18.5 DTC Setting Procedure

Before using the DTC, set the DTC Vector Base Register (DTCVBR). When used in secure state, access DTCVBR_SEC instead of DTCVBR. Set the ICU.IELSR n .IELS[8:0] bits to 0 to disable the interrupt in the NVIC and follow the procedure in [Table 18.8](#) to set the DTC.

Table 18.8 DTC setting procedure

No.	Step Name	Description
1	Set the DTCCR.RRS bit to 0	Set the DTCCR.RRS bit to 0 to reset the transfer information read skip flag. After that, the transfer information read is not skipped while the DTC is activated. Be sure to specify this setting when the transfer information is updated.
2	Set transfer information (MRA, MRB, MRC, SAR, DAR, CRA, and CRB)	Allocate transfer information (MRA, MRB, MRC, SAR, DAR, CRA, and CRB) in the data area. To set transfer information, see section 18.2, Register Descriptions . To allocate transfer information, see section 18.3.1, Allocating Transfer Information and DTC Vector Table .
3	Set transfer information start addresses in the DTC vector table	Set the transfer information start addresses in the DTC vector table. To set the DTC vector table, see section 18.3.1, Allocating Transfer Information and DTC Vector Table .
4	Set the DTCCR.RRS bit to 1	Set the DTCCR.RRS bit to 1 to enable skipping of the second and subsequent transfer information read cycles for continuous DTC activation from the same interrupt source. The RRS bit can be set to 1, but if this is set during DTC transfer, it becomes valid from the next transfer. Leave the value of the RRS bit as 0 if the value of the MRA.WBDIS bit or the MRC.DISPE bit in any transfer information is 1.
5	Set the ICU.IELSR n .DTCE bit to 1. Set the ICU.IELSR n .IELS[8:0] as interrupt source. The interrupt should be enabled in the NVIC.	Set the ICU.IELSR n .DTCE bit to 1. Set ICU.IELSR n .IELS[8:0] as interrupt sources that trigger DTC. The interrupt must be enabled in the NVIC. See section 14.3.2, Event Number in section 14, Interrupt Controller Unit (ICU) .
6	Set the enable bit for an activation source interrupt	Set the enable bit for the activation source interrupts to 1. When a source interrupt is generated, the DTC is activated. To set the interrupt source enable bit, see the settings for the modules that are to be the activation sources.
7	Set the DTCST.DTCST bit to 1	Set the DTC Module Start bit (DTCST.DTCST) to 1.

Note: The DTCST.DTCST bit can be set even if the setting for each activation source is not completed.

Note: When DTCSAR.DTCSTSA = 0 is set and DTC is used in both the Secure and the Non-secure state, DTCST.DTCST bit must be written in the Secure state.

Note: When used in secure state for step 1 and step 4, access DTCCR_SEC.RRS instead of DTCCR.RRS.

18.6 Examples of DTC Usage

18.6.1 Normal Transfer

This section provides an example of DTC usage and its application when receiving 128 bytes of data from an SCI.

(1) Transfer information settings

In the MRA register, select a fixed source address (MRA.SM[1:0] = 00b), normal transfer mode (MRA.MD[1:0] = 00b), and byte-sized transfer (MRA.SZ[1:0] = 00b). In the MRB register, specify incrementation of the destination address (MRB.DM[1:0] = 10b) and single data transfer by a single interrupt (MRB.CHNE = 0 and MRB.DISEL = 0). The MRB.DTS bit can be set to any value. Set the RDR register address of the SCI in the SAR register, the start address of the SRAM area for data storage in the DAR register, and 128 (0x0080) in the CRA register. The CRB register can be set to any value.

(2) DTC vector table settings

The start address of the transfer information for the RXI interrupt is set in the vector table for the DTC.

(3) ICU settings and DTC module activation

Set the ICU.IELSRn.DTCE bit to 1 and set ICU.IELSRn.IELS[9:0] as the SCI interrupt. The interrupt must be enabled in the NVIC. Set the DTCST.DTCST bit to 1.

(4) SCI settings

Enable the SCIn_RXI (n = 0 to 9) interrupt by setting the SCR.RIE bit in the SCI to 1. If a reception error occurs during the SCI receive operation, reception stops. To manage this, use settings that allow the CPU to accept receive error interrupts.

(5) DTC transfer

Each time a reception of 1 byte by the SCI is complete, an SCIn_RXI interrupt is generated to activate the DTC. The DTC transfers the received byte from the RDR of the SCI to the SRAM, after which the DAR register is incremented and the CRA register is decremented.

(6) Interrupt handling

After 128 rounds of data transfer are complete and the value in the CRA register becomes 0, an SCIn_RXI interrupt request is generated for the CPU. Complete the process in the handling routine for this interrupt.

18.6.2 Chain Transfer

This section provides an example of chain transfer by the DTC and describes its use in the output of pulses by the General PWM Timer (GPT). You can use chain transfer to transfer PWM timer compare data and change the period of the PWM timer for the GPT.

For the first of the chain transfers, normal transfer mode is specified for transfer to the GPT32n.GTCCRC register (n = 0 to 13). For the second of the chain transfers, normal transfer mode is specified for transfer to the GPT32n.GTCCRE register (n = 0 to 13). For the third of the chain transfers, normal transfer mode is specified for transfer to the GPT32n.GTPBR register (n = 0 to 13). This is because clearing of the activation source and generation of an interrupt on completion of the specified number of transfers are restricted to the third of the chain transfers, that is, transfer while MRB.CHNE = 0.

The following example shows how to use the counter overflow interrupt with the GPT320.GTPR register as an activating source for the DTC.

(1) First transfer information set

Set up transfer to the GPT320.GTCCRC register.

1. In the MRA register, select incrementation of the source address (MRA.SM[1:0] = 10b).
2. Set the transfer to normal transfer mode (MRA.MD[1:0] = 00b) and word-sized transfer (MRA.SZ[1:0] = 10b).
3. In the MRB register, select the destination address as fixed (MRB.DM[1:0] = 00b) and set up chain transfer (MRB.CHNE = 1 and MRB.CHNS = 0).
4. Set the SAR register to the first address of the data table.

5. Set the DAR register to the address of the GPT320.GTCCRC register.
6. Set the CRA register to the size of the data table.

(2) Second transfer information set

Set up for transfer to the GPT320.GTCCRE register.

1. In the MRA register, select incrementation of the source address (MRA.SM[1:0] = 10b).
2. Set the transfer to normal transfer mode (MRA.MD[1:0] = 00b) and word-sized transfer (MRA.SZ[1:0] = 10b).
3. In the MRB register, select the destination address as fixed (MRB.DM[1:0] = 00b) and set up chain transfer (MRB.CHNE = 1, MRB.CHNS = 0).
4. Set the SAR register to the first address of the data table.
5. Set the DAR register to the address of the GPT320.GTCCRE register.
6. Set the CRA register to the size of the data table.

(3) Third transfer information set

Set up transfer to the GPT320.GTPBR register.

1. In the MRA register, select incrementation of the source address (MRA.SM[1:0] = 10b).
2. Set the transfer to normal transfer mode (MRA.MD[1:0] = 00b) and word-sized transfer (MRA.SZ[1:0] = 10b).
3. In the MRB register, select the destination address as fixed (MRB.DM[1:0] = 00b) and end chain transfer (MRB.CHNE = 0, MRB.CHNS = 0, MRB.DISEL = 0).
4. Set the SAR register to the first address of the data table.
5. Set the DAR register to the address of the GPT320.GTPBR register.
6. Set the CRA register to the size of the data table.

(4) Transfer information assignment

Place the transfer information for use in the transfer to the GPT320.GTPBR immediately after the transfer control information for use in the GPT320.GTCCRC and GPT320.GTCCRE registers.

(5) DTC vector table

In the DTC vector table, set the address where the transfer control information for use in transfer to the GPT320.GTCCRC and GPT320.GTCCRE registers starts.

(6) ICU setting and DTC module activation

1. Set the ICU.IELSRn.DTCE bit associated with the GPT320 counter overflow interrupt.
2. Set the ICU.IELSRn.IELS[9:0] bits and specify the GPT320 counter overflow.
3. Set the DTCST.DTCST bit to 1.

(7) GPT settings

1. Set the GPT320.GTIOR register so that the GTCCRA and GTCCRB registers operate as output compare registers.
2. Set the default PWM timer compare values in the GPT320.GTCCRA and GPT320.GTCCRB registers and the next PWM timer compare values in the GPT320.GTCCRC and GPT320.GTCCRE registers.
3. Set the default PWM timer period values in the GPT320.GTPR register and the next PWM timer period values in the GPT320.GTPBR register.
4. Set 1 to the output bit in PmnPFS.PDR, and set 00011b to the Peripheral Select bits in PmnPFS.PSEL[4:0].

(8) GPT activation

Set the GPT320.GTSTR.CSTRT bits to 1 to start the GPT320.GTCNT counter.

(9) DTC transfer

Each time a GPT320 counter overflow is generated with the GPT320.GTPR register, the next PWM timer compare values are transferred to the GPT320.GTCCRC and GPT320.GTCCRE registers. The setting for the next PWM timer period is transferred to the GPT320.GTPBR register.

(10) Interrupt handling

After the specified rounds of data transfer are complete, for example when the value in the CRA register for GPT transfer becomes 0, a GPT320 counter overflow interrupt request is issued for the CPU. Complete the process for this interrupt in the handling routine.

18.6.3 Chain Transfer when Counter = 0

The second data transfer is performed only when the transfer counter is set to 0 in the first data transfer, and the first data transfer information is repeatedly changed in the second transfer. Chain transfer enables transfers to be repeated 256 times or more.

The following procedure shows an example of configuring a 1-KB input buffer, where the input buffer is set so that its lower address starts with 0x00. [Figure 18.14](#) shows a chain transfer when the counter = 0.

1. Set the normal transfer mode to input data for the first data transfer. Set the following:
 - (a) Transfer source address = fixed.
 - (b) CRA register = 0x0200 (512) times.
 - (c) MRB.CHNE bit = 1 (chain transfer is enabled).
 - (d) MRB.CHNS bit = 1 (chain transfer is performed only when the transfer counter is 0).
 - (e) MRB.DISEL bit = 0 (an interrupt request to the CPU is generated when the specified data transfer completes).
2. Prepare the upper 8-bit address of the start address at every 512 times of the transfer destination address for the first data transfer in different area such as the flash. For example, when setting the input buffer to 0x8000 to 0x83FF, prepare 0x82 and 0x80.
3. For the second data transfer:
 - (a) Set the repeat transfer mode (with transfer source and destination address = fixed.) to reset the transfer counter of the first data transfer.
 - (b) Specify the CRA register in the first transfer information area for the transfer destination.
 - (c) Set the MRB.CHNE bit = 1 (chain transfer is enabled).
 - (d) Set the MRB.CHNS bit = 0 (select continuous chain transfer).
 - (e) Set the MRB.DISEL bit = 0 (an interrupt request to the CPU is generated when the specified data transfer completes).
 - (f) CRA register = 0x0101 (The transfer count is 1).
4. For the third data transfer:
 - (a) Set the repeat transfer mode (with the source as the repeat area) to reset the transfer destination address of the first data transfer.
 - (b) Specify the upper 8 bits of the DAR register in the first transfer information area for the transfer destination.
 - (c) Set the MRB.CHNE bit = 0 (chain transfer is disabled).
 - (d) Set the MRB.DISEL bit = 0 (an interrupt request to the CPU is generated when the specified data transfer completes).
 - (e) When setting the input buffer to 0x8000 to 0x83FF, also set the transfer counter to 2.
5. The first data transfer is performed by an interrupt 512 times. When the transfer counter of the first data transfer becomes 0, the second data transfer starts. Set the transfer counter of the first data transfer to 0x0200. The lower 8 bits of the transfer destination address and the transfer counter of the first data transfer becomes 0x0200.
6. The second data transfer is performed by an interrupt 1 times. When the transfer counter of the first data transfer becomes 0, the third data transfer starts. Set the upper 8 bits of the transfer destination address of the first data transfer to

0x82. The lower 8 bits of the transfer destination address becomes 0x00 and the transfer counter of the first data transfer becomes 0x0200.

7. In succession, the first data transfer is performed by an interrupt 512 times as specified for the first data transfer. When the transfer counter of the first data transfer becomes 0, the second data transfer starts. Set the transfer counter of the first data transfer to 0x0200. The lower 8 bits of the transfer destination address and the transfer counter of the first data transfer becomes 0x0200.
8. The second data transfer is performed by an interrupt 1 times. When the transfer counter of the first data transfer becomes 0, the third data transfer starts. Set the upper 8 bits of the transfer destination address of the first data transfer to 0x80. The lower 8 bits of the transfer destination address becomes 0x00 and the transfer counter of the first data transfer becomes 0x0200.
9. Steps 5 to 8 are repeated indefinitely. Because the second data transfer is in repeat transfer mode, no interrupt request to the CPU is generated.

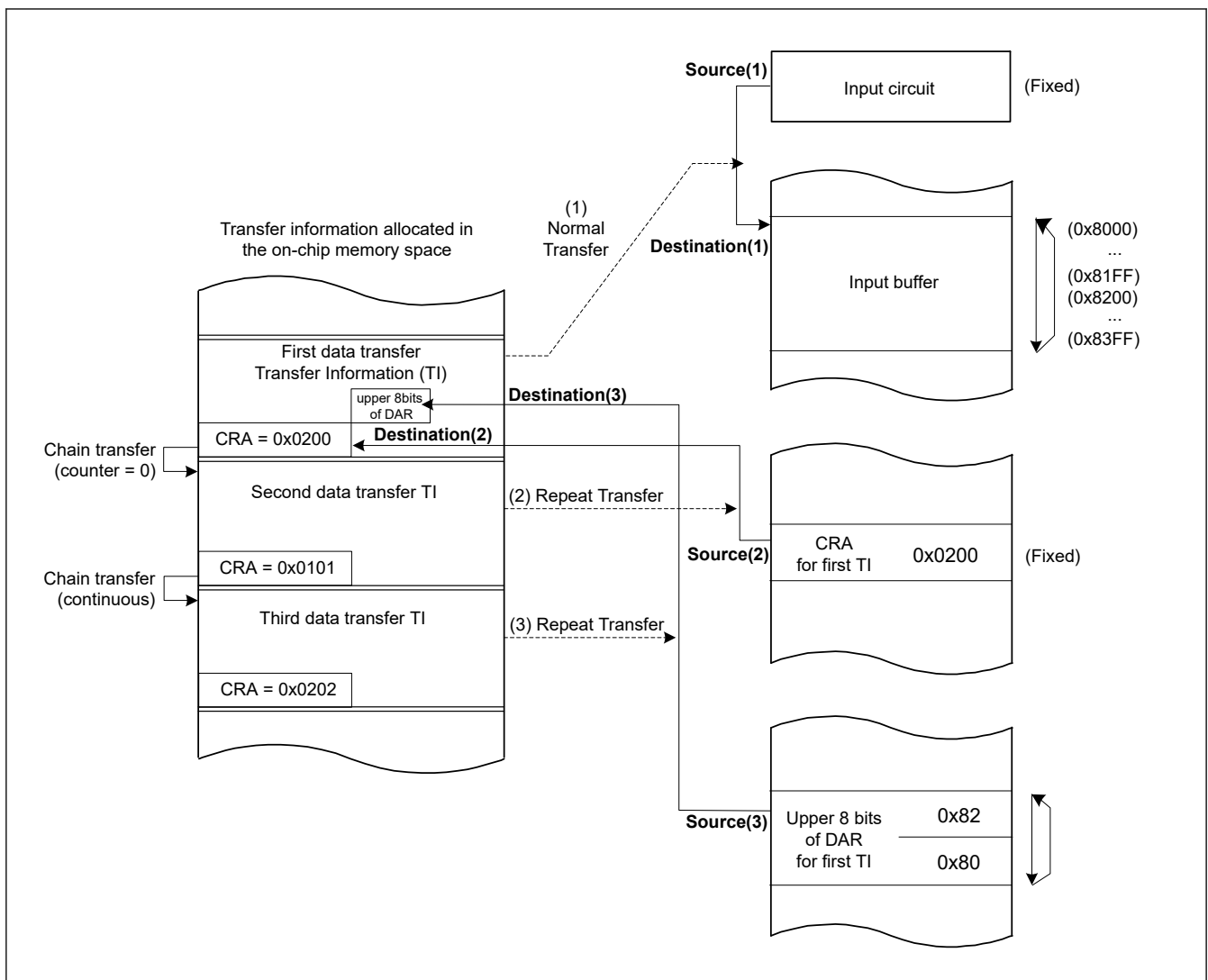


Figure 18.14 Chain transfer when counter = 0

18.7 Processing on DTC Transfer Error

If the access error occurs during DTC transfer, the DTC immediately stops access during transfer. DTC is stopped only for the vector number that caused the error.

The error response is informed to ICU when the transfer error occurs. ICU clears the ICU.IELSRn of the corresponding vector number which caused the transfer error. Write back to SRAM is not performed.

After that, if there is a request other than the vector number which caused the error, it will be re-arbitration as it is.

18.8 Interrupt

18.8.1 Interrupt Request of Transfer End

When the DTC completes data transfer of the specified count or when data transfer with MRB.DISEL set to 1 is complete, a DTC activation source generates an interrupt to the CPU. Interrupts to the CPU are controlled according to the settings in the NVIC and the ICU.IELSRn.IELS[9:0] bits. See [section 14, Interrupt Controller Unit \(ICU\)](#). The DTC prioritizes activation sources by granting the smaller interrupt vector numbers higher priority. The priority of interrupts to the CPU is determined by the NVIC priority.

18.8.2 Interrupt Request of Transfer Error

The error response detection interrupt request DMAm_TRANSERR (m = 0, 1) is generated from the DMAC/DTC when the transfer error is detected during DTC transfer.

The types of interrupts that occur when the DTC transfer error occurs are listed in [Table 18.9](#). The [Table 18.9](#) also shows error information stored when a transfer error occurs.

Table 18.9 Interrupt and error information due to DTC transfer error cause

Transfer error factor	NMI/RESET*1 Request	Interrupt Request	Bus Error Status	Error Channel Information
MSAU Error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.MESRRSTAT	DTC.DTEVR
Master MPU Error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.MMERRSTAT	DTC.DTEVR
Illegal Access Error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.ILERRSTAT	DTC.DTEVR
Slave Bus Error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.SLERRSTAT	DTC.DTEVR
TrustZone Filter error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.SLERRSTAT	DTC.DTEVR

Note 1. If ICU.NMIER.BUSEN is enabled with DMA0_TRANSERR set in ICU.IELSR, NMI and an interrupt will occur due to a transfer error caused by DMA. Only NMI can be generated by not setting DMAm_TRANSERR to ICU.IELSR.

18.9 Event Link

The DTC can produce an event link request on completion of one transfer request. The event link request is generated when the bus responds to the last destination access of DTC transfer. The timing of the bus response to DTC write access depends on the bus configuration and transfer destination. For details on bus responses, see [section 15.2.5. Restrictions](#). Note that all DTC transfers are non-bufferable.

18.10 Low Power Consumption Function

Before transitioning to the module-stop state, Software Standby mode or Deep Software Standby mode, set the DTCST.DTCST bit to 0, and then perform the operations described in the following sections.

(1) Module-stop function

Setting the module-stop bit to 1 stops the DTC corresponding to that module-stop bit.

If the module-stop bit is set to 1 while DTC transfers data, the DTC corresponding to that bit becomes the module-stop state after completing it.

When the module-stop bit is 1, access to the register of the DTC corresponding to that module-stop bit is prohibited.

(2) Software Standby mode and Deep Software Standby mode

Use the settings described in [section 11.6.2.1. Entering Software Standby Mode](#) or [section 11.6.3.1. Transition to Deep Software Standby Mode](#).

If DTC transfer operations are in progress when the WFI instruction is executed, the transition to Software Standby mode or Deep Software Standby mode follows the completion of the DTC transfer.

(3) Notes on Low Power Consumption Function

For the WFI instruction and the register setting procedure, see [section 11, Low Power Mode](#).

To perform a DTC transfer after returning from a low power mode.

To use a request that is generated in Software Standby mode as an interrupt request to the CPU but not as a DTC activation request, specify the CPU as the interrupt request destination as described in [section 14.4.1. Interrupt Detection Selection](#), then execute the WFI instruction.

18.11 Usage Notes

18.11.1 Transfer Information Start Address

You must set multiples of 16 for the transfer information start addresses in the vector table.

18.11.2 Arrangement of Transfer Information

When allocating the transfer information to the memory, arrange it as shown in [Figure 18.15](#).

For example, when writing CRA and CRB setting data in 16 bits, write the CRB setting data to the lower address 0 and the CRA setting data to the lower address 2. When writing in 32 bits, place the CRA setting data on the 32-bit MSB side and the CRB setting data on the LSB side, and write them to the lower address 0.

Also, arrange the transfer information so that it does not straddle the 4KB boundary.

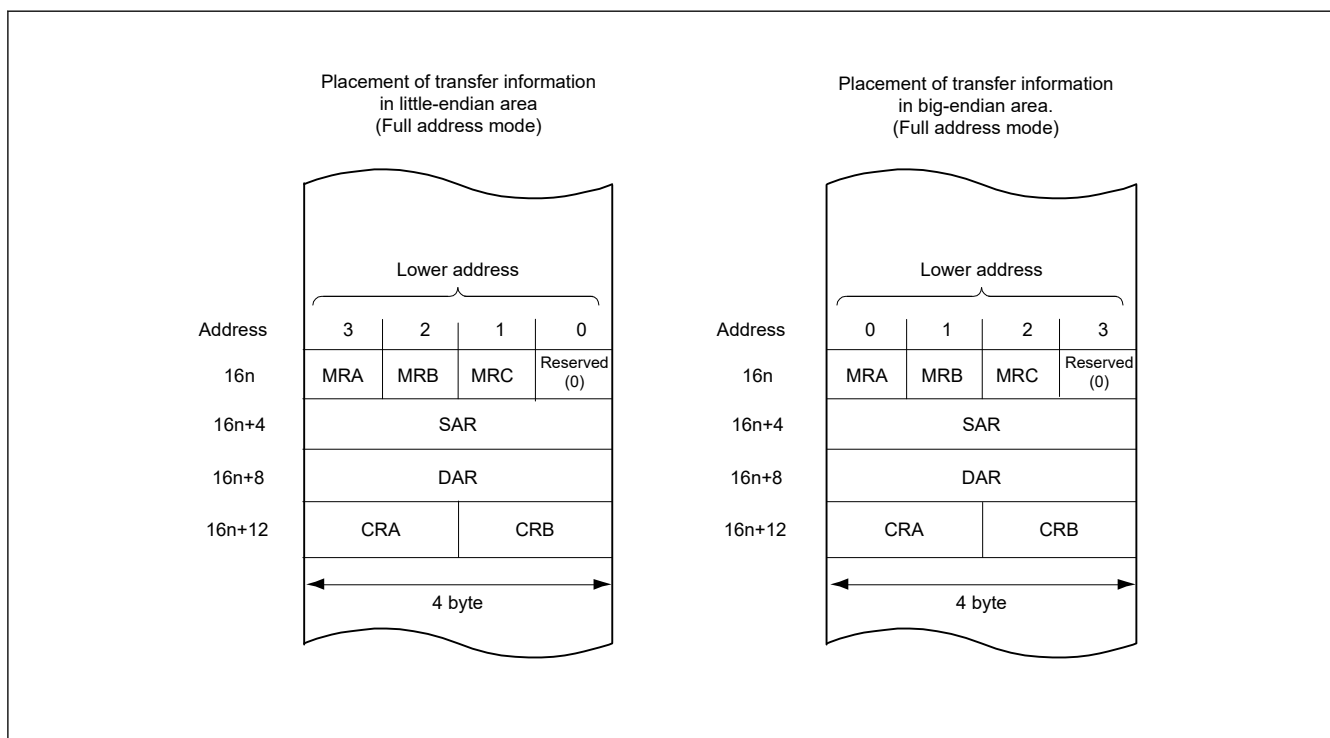


Figure 18.15 Arrangement of transfer information

18.11.3 Bus Response to DTC Write Access

The timing of the bus response to DTC write access depends on the transfer destination.

When writing to an access destination described in [section 15.2.5.3. Notes on Writing Completion Timing](#), a bus response is returned before the write operation is completed.